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ROTATIONAL ATHERECTOMY DEVICES AND METHODS

Abstract

Some embodiments of a rotational atherectomy device can remove (partially or completely) stenotic lesions in blood vessels by rotating one or more abrasive elements in an orbital path to abrade and breakdown the lesion. In particular implementations, multiple abrasive elements are arranged along a distal portion of a drive shaft with an improved configuration so as to facilitate both efficient navigation into smaller blood vessels below the ankle or in the heart and effective orbital paths for abrading stenotic material in such smaller vessels.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/774,078, filed on Jul. 16, 2024, which is a continuation of U.S. application Ser. No. 18/658,372 filed on May 8, 2024 (now U.S. Pat. No. 12,193,702), which is a divisional of U.S. application Ser. No. 18/523,683 filed on Nov. 29, 2023 (now U.S. Pat. No. 12,004,771), which claims priority under 35 U.S.C. § 119 to U.S. Provisional Application Ser. No. 63/542,438, filed on Oct. 4, 2023, and to U.S. Provisional Application Ser. No. 63/523,583, filed on Jun. 27, 2023, the contents of each are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] This document relates to rotational atherectomy devices and systems for removing or reducing stenotic lesions in blood vessels, for example, by urging one or more abrasive elements in an orbital motion within the vessel to remove (partially or completely) the stenotic lesion material.

BACKGROUND

[0003] Atherosclerosis, the clogging of arteries with plaque, is often a result of coronary heart disease or vascular problems in other regions of the body. Plaque can be formed from fat, cholesterol, calcium, and other substances found in the blood. Over time, the plaque hardens and narrows the arteries. This limits the flow of oxygen-rich blood to organs and other parts of the body.

[0004] Blood flow through the central and peripheral arteries (e.g., carotid, iliac, femoral, renal, etc.) can be similarly affected by the development of atherosclerotic blockages. For example, peripheral artery disease (PAD) can be serious because without adequate blood flow, the kidneys, legs, arms, and feet may suffer irreversible damage. Left untreated, the tissue may die or harbor infection. In another example, coronary artery disease (CAD) arises from the buildup of atherosclerotic material in one or more coronary arteries and may result in a deprivation of blood, oxygen, and nutrients to the heart muscle.

[0005] Rotational atherectomy can be used to treat such blockages in some types of blood vessels. In some versions of rotational atherectomy, a drive shaft carrying an abrasive burr or other abrasive surface (e.g., having diamond grit or diamond particles) rotates at a high speed within the vessel, and the clinician operator slowly advances the atherectomy device distally so that the abrasive burr scrapes against the occluding lesion and grinds it into very small particles, reducing the occlusion and improving blood flow through the vessel. While rotational atherectomy is commonly performed in larger arteries in the leg, such a process can be obstructed in small arteries, such as those below the ankle (e.g., in the feet) or in particular coronary arteries (e.g., in the left anterior descending coronary artery or the left circumflex coronary artery), especially where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route.

SUMMARY

[0006] Some embodiments of rotational atherectomy systems described herein can remove (partially or completely) stenotic lesions in blood vessels by rotating one or more abrasive elements in an orbital path to abrade and breakdown the lesion. In particular implementations, multiple abrasive elements are arranged along a distal portion of a drive shaft with an improved configuration/relative sizing so as to facilitate both efficient navigation into smaller blood vessels (below the ankle or in the heart) and effective orbital paths for abrading stenotic material in such

smaller vessels. Additionally, some versions of the improved configuration of the multiple abrasive elements are arranged along the drive shaft can be effective for the removal or reduction of stenotic lesions in larger vessels too (e.g., those in the leg above the ankle or larger coronary arteries), thereby providing the user with options for efficiently treating a variety of arterial sites during a single procedure.

[0007] In one aspect, this disclosure is directed to a rotational atherectomy device for removing stenotic lesion material from a pedal artery or coronary artery of a patient. The device can include an elongate flexible drive shaft defining a longitudinal axis and comprising a torque-transmitting coil of one or more filars that are helically wound around the longitudinal axis in a filar wind direction from a distal end to a proximal end. The device may also include one or more abrasive burrs (e.g., such as a series of abrasive burrs as illustrated herein below) fixedly mounted to a distal end portion of the torque-transmitting coil. Optionally, the abrasive burrs can have a maximum burr diameter of no greater than 1.25 mm. At least one of the abrasive burrs can be an eccentric abrasive burr having a center of mass offset from the longitudinal axis so as to orbit in a rotational direction opposite of the filar wind direction.

[0008] In another aspect, this disclosure describes a method for rotational atherectomy in a pedal artery below an ankle. The method can include advancing a torque-transmitting coil of a rotation atherectomy device over a guidewire and into a pedal artery below an ankle so that at least one abrasive burr mounted to a distal end portion of the torque-transmitting coil is proximate to a stenotic lesion within the pedal artery. Additionally, the method can further include rotating the torque-transmitting coil of the rotation atherectomy device so that the at least one abrasive burr mounted to the torque-transmitting coil abrades the stenotic lesion within the pedal artery.

[0009] Another aspect set forth in this disclosure is a method for rotational atherectomy in a coronary branch artery of a heart. The method can include advancing a torque-transmitting coil of a rotation atherectomy device over a guidewire and into a coronary branch artery of a heart so that at least one abrasive burr mounted to a distal end portion of the torque-transmitting coil is proximate to a stenotic lesion within the coronary branch artery. The method may also include rotating the torque-transmitting coil of the rotation atherectomy device so that the at least one abrasive burr mounted to the torque-transmitting coil abrades the stenotic lesion within the coronary branch artery.

[0010] Further aspects set forth in this disclosure include at least one of a 4-French introducible rotational atherectomy device, a pedal loop navigable rotational atherectomy device, and a coronary branch navigable rotational atherectomy device (as detailed below).

[0011] In another aspect, this disclosure describes a rotational atherectomy system. The system can include a torque-transmitting coil of one or more filars that are helically wound around in a filar wind direction from a distal end to a proximal end to define a coil diameter and a drive shaft axis. The system can further include one or more abrasive burrs (e.g., such as a series of abrasive burrs as illustrated herein below) fixedly mounted to a distal end portion of the torque-transmitting coil. Optionally, each of the abrasive burrs can have an outer burr diameter such that a burr-to-coil diameter ratio for all abrasive burrs along the torque-transmitting coil is 1.3 to 1.7. The system may also include a rotational atherectomy handle assembly coupled to a proximal end of the torque-transmitting coil. Optionally, the rotational atherectomy handle assembly may house an electric motor configured to, responsive to user input at an actuator of the rotational atherectomy handle assembly, drive rotation of the abrasive burrs about the drive shaft axis in a rotational direction.

[0012] In yet another aspect, this disclosure is directed to a rotational atherectomy device for removing stenotic lesion material from an artery. The device can include an elongate flexible drive shaft defining a longitudinal axis and comprising a torque-transmitting coil of one or more filars that are helically wound around the longitudinal axis in a filar wind direction from a distal end to a proximal end. The device can also include a series of abrasive burrs fixedly mounted to a distal end portion of the torque-transmitting coil and having a maximum burr diameter of no greater than 1.25

mm. Optionally, an intermediate abrasive burr of the series of abrasive burrs is an eccentric abrasive burr having a center of mass offset from the longitudinal axis so as to orbit in a rotational direction opposite of the filar wind direction. The series of abrasive burrs can also include a proximal abrasive burr and a distal abrasive burr that each have a center of mass coaxial with the longitudinal axis.

[0013] Another aspect set forth in this disclosure is a rotational atherectomy system that includes a torque-transmitting coil, a series of abrasive burrs, and a rotational atherectomy handle assembly. The torque-transmitting coil can include one or more filars that are helically wound around in a filar wind direction from a distal end to a proximal end to define a coil diameter and a drive shaft axis. The series of abrasive burrs can be fixedly mounted to a distal end portion of the torque-transmitting coil and, optionally, may have a maximum burr diameter of no greater than 1.25 mm. The series of abrasive burrs can include a distal concentric abrasive burr, a proximal concentric abrasive burr, and an intermediate eccentric abrasive burr. Optionally, the distal concentric abrasive burr is coaxially mounted to a distal-most end of the torque-transmitting coil, the proximal concentric abrasive burr has the same size as the distal concentric abrasive burr and is coaxially mounted to the torque-transmitting coil at a position no greater than 0.5 inches from the distal-most end of the torque-transmitting coil, and the intermediate eccentric abrasive burr has a larger size than the distal and proximal concentric abrasive burrs and is mounted to the torque-transmitting coil with a center of mass offset from the drive shaft axis. Additionally, the rotational atherectomy handle assembly can be coupled to a proximal end of the torque-transmitting coil and may house an electric motor configured to, responsive to user input at an actuator of the rotational atherectomy handle assembly, drive rotation of the series of abrasive burrs about the drive shaft axis in a rotational direction.

[0014] Some of the embodiments described herein may provide one or more of the following advantages. First, some embodiments of the rotational atherectomy system can be configured to provide a rotational atherectomy treatment in small arteries, such as those below the ankle (e.g., in the feet) or coronary arteries (e.g., in the left anterior descending coronary artery or the left circumflex coronary artery). In some examples described below, the orientation, relative spacing, and relative size of the abrasive elements along the distal end portion of the drive shaft (along with other features of the drive shaft) can collectively achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route.

[0015] Second, some embodiments of the rotational atherectomy devices and systems provided herein can advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath, and can have a sufficient balance of factors (e.g., length, maximum lateral radius from a central axis, flexibility, torque-transmission capabilities, etc.) to advance into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery. In particular versions described below, the rotational atherectomy device can be in the form of a "4-French introducer" device, which as used herein means a rotational atherectomy device in which a maximum radius (measured from a central longitudinal axis of the torque-transmitting coil) among all of the torque-transmitting coil and the abrasive elements positioned thereon is smaller than the central access path of a 4-French introducer sheath, thereby providing slidable insertion through the 4-French introducer sheath.

[0016] Third, some embodiments of the rotational atherectomy devices and systems described herein have a relative sizing ratio between a diameter of each abrasive element and a coil diameter of the drive shaft that is 1.7 or less. In particular examples described below, this relative sizing ratio is referred to as the burr-to-coil diameter ratio, and the burr-to-coil diameter ratio can be selected in accordance with the teachings herein in order to advantageously provide a drive shaft that can advance through a stenotic lesion in a small artery and then subsequently achieve an orbital path during rotation (e.g., an orbital path that is greater in size than a maximum stationary diameter of

the abrasive elements) to abrade the stenotic lesion. In some implementations, the burr-to-coil diameter ratio can be about 1.3-1.7 for all abrasive elements along the torque-transmitting coil of the drive shaft. For example, the burr-to-coil diameter ratio can be 1.5-1.6 for a largest abrasive element mounted to the torque-transmitting coil and 1.3-1.4 for a smallest abrasive element mounted to the torque-transmitting coil. In a further example, the burr-to-coil diameter ratio can be 1.5-1.6 for each eccentric abrasive element mounted to the torque-transmitting coil and 1.3-1.4 for each concentric abrasive element mounted to the torque-transmitting coil.

[0017] Fourth, in some embodiments of the rotational atherectomy devices and systems that include multiple abrasive elements (at least one of which has a center of mass offset from the central axis of the drive shaft) that are longitudinally spaced apart for an advantageously condensed length of less than 5 cm from the distal-most tip of the drive shaft (preferably 2 cm or less from the distal-most tip of the drive shaft), with the combined radial angles of the all abrasive elements differing by less than 120 degrees (e.g., less than 90 degrees for the depicted embodiments herein) along that particular length of the drive shaft. Such a design can advantageously assist the eccentric abrasive elements to advance into tortuous arterial paths and small arteries and then (during rotation) achieve an orbit path that abrades targeted stenotic material from the arterial wall.

[0018] Fifth, some embodiments of the rotational atherectomy devices and systems provided herein can include an improved control handle that achieves simplified setup and convenient operation for a clinician. In particular examples, that the control handle can be a one-time-use, disposable unit that houses both an electric motor (for driving rotation of the drive shaft) and a fluid pump (for delivery of saline or another fluid toward the distal end of the drive shaft). Optionally, the controller for the handle (which includes a processor and memory storing the control instructions) can be housed in a separate, screenless housing (e.g., with the power adapter unit that plugs into an ordinary wall socket). In such optional implementations, the controller does not require a user interface screen and instead the user interface buttons are provided along the control handle that is connected to the controller via a detachable cable. Accordingly, in these optional implementations, the controller can be reusable over time with multiple control handles (all of which are one-time-use, disposable unit), thereby advantageously conserving costs while preserving the convenient disposability of the driveshaft and handle.

[0019] Sixth, some embodiments of the rotational atherectomy devices and systems provided herein can be advantageously in the form of a “pedal loop navigable” device, which as used herein means a rotational atherectomy device in which the torque-transmitting coil provides flexibility during advancement through the tortuous path into a pedal loop of a foot below the ankle and the abrasive elements are oriented and sized to advance through a stenotic lesion within the foot for rotational atherectomy treatment in the foot.

[0020] Seventh, some embodiments of the rotational atherectomy devices and systems provided herein can be advantageously in the form of a “coronary branch navigable” device, which as used herein means a rotational atherectomy device in which the torque-transmitting coil provides flexibility during advancement through the tortuous path into a branch coronary artery (e.g., the left anterior descending coronary artery or the left circumflex coronary artery) of a heart and the abrasive elements are oriented and sized to advance through a stenotic lesion within the branch coronary artery for rotational atherectomy treatment in the heart.

[0021] Eighth, some embodiments of the rotational atherectomy devices and systems provided herein can include an improved handle and sheath that facilitates retraction and extension of the abrasive elements to and from the sheath. The retraction of the abrasive elements within the sheath advantageously covers the abrasive burrs while the rotational atherectomy device is navigated to small arteries, such as those below the ankle (e.g., in the feet) or coronary arteries (e.g., in the left anterior descending coronary artery or the left circumflex coronary artery). The atherectomy device can be advantageously advanced to the target area while the abrasive surfaces of the abrasive elements are shielded from the vessel walls, thereby reducing the likelihood of scraping or

engaging unintended regions of a bodily vessel along the navigation path. Instead, the atherectomy device is configured to smoothly navigate through the complex anatomy (e.g., especially along smaller arterial paths) while the abrasive elements are in a retracted position within a distal end of the sheath lumen, and then to adjust the abrasive elements distally from the sheath lumen after navigating to the targeted site (e.g., for rotational atherectomy treatment).

[0022] Ninth, some embodiments of the rotational atherectomy devices and systems provided herein can facilitate improved safety and ease of operation of the rotational atherectomy device. For example, the system can be configured to automatically prevent rotation of the drive shaft (and the abrasive elements thereon) when the abrasive elements are in a first longitudinal position (e.g., retracted within the sheath lumen) and to provide selective activation of rotation of the drive shaft when the abrasive elements are in a second longitudinal position (e.g., adjusted to extend distally from the sheath). Accordingly, the system can provide an added safety control that facilitates improved and intuitive operation for a user both during navigation of the drive shaft (when the abrasive elements are positioned within a sheath) and during rotational atherectomy treatment after reaching the targeted site (permitting user-controlled rotation of the abrasive elements after the abrasive elements are extended from the sheath).

[0023] The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

[0024] FIG. 1 a perspective view of an example rotational atherectomy system, in accordance with some embodiments.

[0025] FIG. 2 is a longitudinal side view of a rotational atherectomy device of the system of FIG. 1.

[0026] FIG. 3 is a distal end view of the rotational atherectomy device of FIG. 2.

[0027] FIG. 4 is a perspective view of the distal portion of the example rotational atherectomy system of FIG. 1 in the blood vessel.

[0028] FIG. 5 is a transverse cross-sectional view of a distal tip of the rotational atherectomy device of FIG. 2.

[0029] FIG. 6 is a perspective view of the distal portion of the example rotational atherectomy system of FIG. 1 in a pedal blood vessel below an ankle, in accordance with some embodiments.

[0030] FIG. 7 is a magnified perspective view of the distal portion of the example rotational atherectomy system of FIG. 6 in the pedal blood vessel below the ankle.

[0031] FIG. 8 is a longitudinal side view of an alternative embodiment of a distal portion of a rotational atherectomy device, in accordance with some embodiments.

[0032] FIG. 9 is a distal end view of the rotational atherectomy device of FIG. 8.

[0033] FIG. 10 is a longitudinal side view of an alternative embodiment of a distal portion of a rotational atherectomy device, in accordance with some embodiments.

[0034] FIG. 11 is a distal end view of the rotational atherectomy device of FIG. 10.

[0035] FIG. 12 is a longitudinal side view of an alternative embodiment of a distal portion of a rotational atherectomy device, in accordance with some embodiments.

[0036] FIG. 13 is a distal end view of the rotational atherectomy device of FIG. 12.

[0037] FIG. 14 is a longitudinal side view of an alternative embodiment of a distal portion of a rotational atherectomy device, in accordance with some embodiments.

[0038] FIG. 15 is a distal end view of the rotational atherectomy device of FIG. 14.

[0039] FIG. 16 is a perspective view of the distal portion of the example rotational atherectomy

device of FIG. 14 in a coronary blood vessel, in accordance with some embodiments.

[0040] FIG. 17 is a longitudinal side view of an alternative embodiment of a distal portion of a rotational atherectomy device, in accordance with some embodiments.

[0041] FIG. 18 is a distal end view of the rotational atherectomy device of FIG. 17.

[0042] FIG. 19 is a longitudinal side view of an alternative embodiment of a distal portion of a rotational atherectomy device, in accordance with some embodiments.

[0043] FIG. 20 is a distal end view of the rotational atherectomy device of FIG. 19.

[0044] FIG. 21 is a longitudinal side view of an alternative embodiment of a distal portion of a rotational atherectomy device, in accordance with some embodiments.

[0045] FIG. 22 is a distal end view of the rotational atherectomy device of FIG. 21.

[0046] FIG. 23 is a longitudinal side view of an alternative embodiment of a distal portion of a rotational atherectomy device, in accordance with some embodiments.

[0047] FIG. 24 is a distal end view of the rotational atherectomy device of FIG. 23.

[0048] FIG. 25 is a longitudinal side view of an alternative embodiment of a distal portion of a rotational atherectomy device, in accordance with some embodiments.

[0049] FIG. 26 is a distal end view of the rotational atherectomy device of FIG. 25.

[0050] FIG. 27 is a perspective view of an example rotational atherectomy system, in accordance with some embodiments.

[0051] FIG. 28 is a longitudinal side view of the rotational atherectomy system of FIG. 27 with the rotational atherectomy device in an exemplary proximal position, in accordance with some embodiments.

[0052] FIG. 29 is a longitudinal side view of a distal end of the rotational atherectomy system and rotational atherectomy device of FIG. 28.

[0053] FIG. 30 is a longitudinal side view of the rotational atherectomy system of FIG. 27 with the rotational atherectomy device in an exemplary proximal operating position, in accordance with some embodiments.

[0054] FIG. 31 is a longitudinal side view of a distal end of the rotational atherectomy system and rotational atherectomy device of FIG. 30.

[0055] FIG. 32 is a longitudinal side view of the rotational atherectomy system of FIG. 27 with the rotational atherectomy device in an exemplary distal operating position, in accordance with some embodiments.

[0056] FIG. 33 is a longitudinal side view of a distal end of the rotational atherectomy system and rotational atherectomy device of FIG. 32.

[0057] FIG. 34 is a detailed sectional view area 34 of the rotational atherectomy system from FIG. 30.

[0058] FIG. 35 is a detailed sectional view area 34 of the rotational atherectomy system from FIG. 28.

[0059] Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0060] Referring to FIG. 1, in some embodiments a rotational atherectomy system 100 for removing (partially or completely) a stenotic lesion 107 from a targeted blood vessel 105 can include an actuator handle assembly 110 that controls movement of an elongate flexible drive shaft assembly 130. The drive shaft assembly 130 includes a flexible drive shaft 136, and a distal end portion of the driveshaft 136 includes one or more abrasive elements 140a-c configured to abrade the stenotic lesion 107 in the targeted vessel 105. As described in more detail below, the abrasive elements 140a-c can have a selected configuration and relative sizing along the distal end portion of the drive shaft 136 so as to improve navigation into smaller blood vessels below the ankle or in the heart while also achieving an effective orbital path for abrading the stenotic material 107 in those vessels. (In the depicted example, the targeted blood vessel 105 has a vessel diameter VD of 3 mm or less, and about 2.5 mm as shown, while the initial path through the stenotic lesion is

significantly smaller than that.) Optionally, the abrasive elements **140** and drive shaft **136** can have a selected configuration and relative sizing (e.g., FIGS. **2-3**, **8-15**, and **17-26**) that advantageously provides advancement through a small percutaneous introducer **108** (e.g., sized to slidably receive instruments of 4-French diameter or smaller) at a percutaneous opening **109** in a patient's leg, and can further navigate through a stenotic lesion **107** in a small artery **105**, such as those below the ankle (e.g., in the feet) or coronary arteries (e.g., in the left anterior descending coronary artery or the left circumflex coronary artery) prior to sweeping a larger orbital path (during rotation of the driveshaft **136**) for abrading the stenotic material **107**.

[0061] The system **100** can also include a power adapter **120** and a fluid source **125** (e.g., a saline bag) connectable to the actuator handle assembly **110**, and the actuator handle assembly **110** can house therein an electric motor **112** (configured to drive rotation of the driveshaft **136**) and fluid pump **114** (configured to urge a fluid such as saline toward the distal end portion of the driveshaft **136**). As described in more detail below, a controller **150** for activating the electric motor **112** and the pump **114** (responsive to inputs at the user interface buttons **116** and **117a-c** of the handle assembly **110**) can be contained inside a housing **122** of the power adapter **120** so that it is reusable with subsequent handle assemblies after the first handle assembly **110** is discarded (a single-use handle assembly). Alternatively, a controller **150A** for operating the electric motor **112** and the pump **114** can be contained within the housing in the handle assembly **110** (in proximity to the electric motor **112** and the pump **114**), and the entire handle assembly **110** can be discarded after a single use with a patient. In both options, the handle assembly **110** can be operated by a clinician using a simplified, screenless interface to perform and control the rotational atherectomy procedure (e.g., without a graphic display screen along the handle assembly or on a separate unit connected to the handle assembly).

[0062] Still referring to FIG. **1**, the elongate flexible drive shaft assembly **130** includes a sheath **132** that extends over a majority length of the flexible drive shaft **136** such that the abrasive elements **140a-c** on the distal end portion of the drive shaft **136** are positioned distally of a distal-most end of the sheath **132**. A proximal end of the sheath **132** is fixed to a distal end of the handle assembly **110**. The flexible drive shaft **136** is slidably and rotatably disposed within a lumen of the sheath **132**. The flexible drive shaft **136** defines a longitudinal lumen in which a guidewire **134** is slidably disposed. The guidewire **134** can extend through the handle assembly **110**, the sheath **132**, and the drive shaft **136** such that a proximal end of the guidewire **134** protrudes proximally from a rear port of a guidewire brake **118** at a proximal end of the handle assembly **110** while a distal end of the guidewire **134** extends distally of a distal-most end of the drive shaft **136**. In this embodiment, the flexible drive shaft **136** includes a torque-transmitting coil of one or more helically wound filars that defines the longitudinal lumen along a central longitudinal axis. The drive shaft **136** is configured to rotate about the longitudinal axis while the sheath **132** remains generally stationary. Hence, during a rotational atherectomy procedure, the sheath **132** and the guidewire **134** are generally stationary while the flexible drive shaft **136** is controllably moved (e.g., rotating about the longitudinal axis and periodically longitudinally translating proximally and/or distally).

[0063] In the depicted embodiment, the exposed distal end portion of the driveshaft **136** includes one or more abrasive elements **140a-c**, a (optional) distal stability element **142**, and a (optional) concentric tip member **144** (refer to FIG. **6**). In the depicted embodiment, the one or more abrasive elements includes a set of three eccentric abrasive elements **140a-c** that are fixedly mounted to an exterior of the torque-transmitting coil of the driveshaft **136** such that a center of mass for each abrasive element **140a-c** is offset from a central longitudinal axis of the torque-transmitting coil. In this embodiment, the distal stability element **142** is concentrically-fixed to an exterior of the torque-transmitting coil of the driveshaft **136** between a distal-most one of the eccentric abrasive elements **140a-c** and the concentric tip member **144**. As such, the center of mass of the distal stability element **142** is aligned with the central axis of the drive shaft **136** while the center of mass

of each abrasive element **140a-c** is offset from the central axis of the drive shaft **136**. The concentric tip member **144** (refer to FIG. 6) is affixed to, and extends distally from, the terminal distal-most end of the torque-transmitting coil. As described in more detail below, the concentric tip member **144** has a smoother surface than the abrasive surfaces of the distal stability element **142** and the eccentric abrasive elements **140a-c**, and the concentric tip member **144** can be configured to provide initial penetration (and, optionally, dilation) through the stenotic lesion **107** in the targeted vessel **105**.

[0064] Still referring to FIG. 1, as the drive shaft **136** is rotated about its longitudinal axis, the eccentric abrasive elements **140a-c** (and the portion of the drive shaft **136** to which the one or more abrasive elements **140a-c** are affixed) will be urged in an orbit path relative to the central axis of the drive shaft **136** (also as described below, for example, in connection with FIG. 5). In general, faster speeds (rpm) of rotation of the drive shaft **136** will result in larger diameters of the orbit (within the limits of the vessel diameter). The orbiting one or more abrasive elements **140a-c** will contact the stenotic lesion **107** to abrade the lesion to a reduced size with each traversal path through the lesion **107** (i.e., small particles of the lesion will be abraded from the lesion).

Depending upon the rotation speed and the surrounding environment within the vessel **105**, the rotating distal stability element **142** can remain generally closer to or at the longitudinal axis of the drive shaft **136** during the rotational atherectomy procedure. In some optional embodiments, two or more distal stability elements **140** are included. As described further below, contemporaneous with the rotation of the drive shaft **136**, the drive shaft **136** can be translated back and forth (distally and proximally) along the longitudinal axis of the drive shaft **136**. Hence, the stenotic lesion **107** can be abraded radially and longitudinally by virtue of the simultaneous translation and orbital rotation of the abrasive elements **140a-c**.

[0065] Additionally, the torque-transmitting coil of the flexible drive shaft **136** is laterally flexible so that the drive shaft **136** can readily advance through a tortuous arterial path (e.g., in the pedal loop or in a coronary branch artery), and so that a portion of the drive shaft **136** at, and adjacent to, the one or more abrasive elements **140** can laterally deflect when acted on by the centrifugal forces resulting from the rotation of the one or more eccentric abrasive elements **140**. In the depicted embodiment, the drive shaft **136** comprises one or more helically wound wires (or filars) that provides a uniform coil diameter that is less than the diameters of all of the abrasive elements **140a-c** and the distal stability element **142**. As described in more detail below, this relative sizing is referred to as the burr-to-coil diameter ratio, and the burr-to-coil diameter ratio can be about 1.3-1.7 for all abrasive burrs (elements **140a-c** and distal stability element **142**) along the torque-transmitting coil of the drive shaft. As such, the torque-transmitting coil of the flexible drive shaft **136** can achieve both sufficient lateral flexibility during navigation through a tortuous path (e.g., in a patient's foot or heart) and sufficient longitudinal rigidity to be pushed through a stenotic lesion (while transmitting torque to rotate the abrasive elements **140a-c**) in a small artery. In some embodiments, the one or more helically wound wires (filars) of the torque-transmitting coil of the flexible drive shaft **136** comprise a metallic material such as, but not limited to, stainless steel (e.g., 316, 316L, or 316LVM), nitinol, titanium, titanium alloys (e.g., titanium beta 3), carbon steel, or another suitable metal or metal alloy. Any suitable number of individual filars can be included to construct the drive shaft **136**. For example, in some embodiments one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, or more than fifteen individual filars can be helically wound among each other to make up the drive shaft **136**. As described further below, the direction in which the filars of the drive shaft **136** are wound is a design feature that can be selected to obtain desirable, advantageous operational characteristics. For example, the drive shaft **136** can be formed using one or more filars that are wound about the shaft's central axis in a wind direction that is opposite from the rotational direction of the drive shaft **136** urged by the handle assembly **110**, which can provide a number of benefits and improved safety during use of the drive shaft **136** with the guidewire **134** in small-sized arteries below the ankle or in the heart.

[0066] Still referring to FIG. 1, the torque-transmitting coil of the drive shaft **136** in this embodiment defines a hollow central core (e.g., referred to as a central lumen of the drive shaft **136**), which can slidably receive the guidewire **134** therein. In some embodiments, the lumen can be used to aspirate particulate or to convey fluids that are beneficial for the atherectomy procedure. In use, the guidewire **134** is advanced to the targeted vessel **105**, and then the drive shaft **136** is advanced over the guidewire **134** in order to reach the targeted vessel **105**. The guidewire **134** has a length sufficient to extend through the entire drive shaft **136** and the entire handle assembly **110**. As such, a proximal end of the guidewire **134** protrudes proximally from the rear port of the guidewire brake **118** at a proximal end of the handle assembly **110** while a distal end of the guidewire **134** extends distally of a distal-most end of the drive shaft **136**.

[0067] In the depicted embodiment, the concentric tip member **144** is welded or otherwise fixed to a distal-most end of the torque-transmitting coil of the drive shaft **136** (e.g., axially distal of the coil), and the distal stability element **142** is welded or otherwise fixed to the distal-most end of the torque-transmitting coil of the drive shaft **136** (e.g., radially outward of the coil). As described in more detail below, the smooth initial surface of the concentric tip member **144** followed by the abrasive surface on the distal stability element **142** can help facilitate the initial expansion and abrasion of a pilot path through the stenotic lesion **107** in the targeted vessel **105**.

[0068] Still referring to FIG. 1, the one or more abrasive elements **140a-c** (each of which may also be referred to as an abrasive burr) can comprise a biocompatible material that is coated with an abrasive media such as diamond grit, diamond particles, silicon carbide, and the like. In the depicted embodiment, the abrasive elements **140a-c** includes a total of three discrete abrasive spheres/cylinders that are spaced apart from each other (and spaced relative to the distal stability element **142**) to facilitate both navigation to, and orbital abrading within, a targeted small artery, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, all three of abrasive elements **140a-c** are spheres having the same diameter and are mounted in an eccentric spiral arrangement (described below in connection with FIGS. 2-3). Other embodiments depicted herein (e.g., refer to FIGS. 9-24) can also be used in accordance with the system of **100** of FIG. 1 to facilitate both navigation to, and orbital abrading within, a targeted small artery, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. As with the distal stability element **142**, the abrasive burrs **140a-c** may be mounted to the exterior of the torque-transmitting coil of the drive shaft **136** using a biocompatible adhesive, high temperature solder, welding, press fitting, and the like. Alternatively, the one or more abrasive elements **140a-c** can be integrally formed as a unitary structure with the filars of the drive shaft **136** (e.g., using filars that are wound in a different pattern to create an axially offset structure, or the like).

[0069] Still referring to FIG. 1, the rotational atherectomy system **100** also includes the actuator handle assembly **110**. The actuator handle assembly **110** includes a housing **111** and an internal carriage assembly (not shown) that translates along an actuator slot **113**. For example, a user can grasp the actuator **116** to urge movement along the actuator slot **113**, which causes the internal carriage assembly to slidably translate along the longitudinal axis of the handle assembly **110**, as indicated by the arrow **115**. In some embodiments the carriage assembly can be translated, without limitation, about 8 cm to about 12 cm, or about 6 cm to about 10 cm, or about 4 cm to about 8 cm, or about 6 cm to about 14 cm. As the carriage assembly is translated in relation to the housing **111**, the drive shaft **136** translates in relation to the sheath **132** in a corresponding manner. As such, the user can reciprocate the distal end portion of the drive shaft **136** in distal and proximal directions relative to the stenotic lesion **107** within the targeted vessel **105**.

[0070] The handle assembly **110** has a cable connection **121** with a power adapter **120** (configured to receive electrical power from a power source **128** such as a wall plug) and fluid line connection **126** with a saline source **125**. The cable **121** can communicate both power and data (e.g., when the controller **150** is housed within the power adapter housing **122**), or alternatively, can communicate

electrical power (e.g., when implementing the version with the controller **150A** that is housed in the handle housing **111**). The cable **121** includes a removable connection jack so that the handle assembly **110** can be readily discarded after a single use and the power adapter **120** can be reused with subsequent handle assemblies. The fluid line connection **126** can include a luer fitting and a flow on-off valve so that a user can removably connect the handle assembly to a pole-mounted saline bag or other fluid source **125** without the need for an external pump mechanism positioned exterior to the handle housing **111**.

[0071] Still referring to FIG. **1**, the actuator **116** of the handle assembly **110** includes a rotational power button that activates the electrical motor **112** (carried by the internal carriage assembly) to drive the rotation of the drive shaft **136**. For example, when the rotational power button of the actuator **116** is depressed, power is supplied to the electric motor **112**, which is coupled to the drive shaft **136** via a set of gears. It should be understood that the rotational atherectomy system **100** is configured to rotate the drive shaft **136** at a high speed of rotation (e.g., 20,000-160,000 rpm) such that the eccentric one or more abrasive elements **140a-c** revolve in an orbital path to thereby contact and remove portions of a target lesion **107** (even those portions of the lesion that are spaced farther from the axis of the drive shaft **136** than the maximum radius of the abrasive elements **140a-c**).

[0072] To operate the handle assembly **110** during a rotational atherectomy procedure, a clinician can grasp the actuator **116** and depress rotational power button (on the actuator **116**) with the same hand. The clinician can move (translate) the actuator **116** along the slot **113** distally and proximally by hand (e.g., back and forth in relation to the housing **111**), while maintaining the rotational power button of the actuator **116** in the depressed state. In that manner, a target lesion **107** can be abraded radially and longitudinally by virtue of the resulting orbital rotation and translation of the abrasive elements **140a-c**.

[0073] To further operate the handle assembly **110** during a rotational atherectomy procedure, a clinician can select a rotational speed using electrical switches **117a** and **117b**. In some cases, the rotational speed can be selected through a set of predefined speeds (e.g., at least two predefined speed settings, such as “low” and “high”) with electrical switch **117a** causing an increase in the speed setting and electrical switch **117b** causing a decrease in the speed setting. Optionally, each of the electrical switches **117a-b** can also include a light indicator. For example, when the electrical switches **117a-b** allow for selection for a “high” and “low” speed, respectively, the electrical switches **117a-b** can each have a single light, such that when a speed is selected, the light corresponding to the selected electrical switch **117a** or **117b** is illuminated to inform a clinician of the selected speed. In some embodiments, the light can shine through electrical switches **117** and **117b**. Alternatively, a light can be positioned proximal electrical switch **117a-b**. As another example, when the electrical switches **117a-b** allow modification of a speed between a range of speeds, the light indicator can be a light bar, such that a number of lights illuminated on the light bar correspond to a selected speed.

[0074] Still referring to FIG. **1**, handle assembly **110** can include a fluid pump switch **117c**, which can activate the internal fluid pump **114** to draw fluid (e.g., saline in this embodiment) from the fluid line **126** and urge the fluid through the sheath **132** toward the distal end portion of the drive shaft **136**. As such, the fluid pump switch **117c** can be used to both initially prime the sheath **132** (and remove air before insertion into the patient) and then selectively activate additional flush fluid through the sheath **132** and into the vessel **105**. In some cases, a first depression of the fluid pump switch **117c** will turn the internal pump **114** on, while a second depression will turn the pump **114** off. In some embodiments, the fluid pump switch **117c** includes a light indicator, such that when the pump is on, a light is illuminated to inform the clinician that the pump is on.

[0075] In the depicted embodiment, the handle assembly **110** also includes a guidewire brake **118** that can be selectively actuated (e.g., pivoted relative to the handle housing **111** in this embodiment) to releasably clamp the guidewire **134** in a stationary position relative to the handle

assembly **110** (and, in turn, stationary in relation to rotations of the drive shaft **136** during an atherectomy treatment). While the drive shaft **136** and handle assembly **110** are being advanced over the guidewire **134** to put the one or more abrasive elements **140** into a targeted position within a patient's vessel, the guidewire brake **118** is in a non-activated state (e.g., pivoted counter-clockwise about the central guidewire axis) from a rear perspective) so that the handle assembly **110** is free to slide in relation to the guidewire **134**. Then, when the clinician is ready to begin the atherectomy treatment, the guidewire brake **118** can be activated (e.g., pivoted clockwise about the central guidewire axis) to mechanically engaged an exterior of the guidewire **134** and thereby releasably detain/lock the guidewire **134** in relation to the handle assembly **110**. That way the guidewire **134** will not rotate while the drive shaft **136** is rotating, and the guidewire **134** will not translate while the actuator **116** is being manually translated in the direction **115**.

[0076] Still referring to FIG. **1**, handle assembly **110** can include a guidewire brake light **119** that positioned along an upper face of the handle housing **111** at a position proximal to the other user interface buttons **117a-c** and adjacent to the guidewire brake **118**. As such, a user can readily view the guidewire brake light **119** and receive confirmation of whether the guidewire brake **118** is fully activated (to clamp the guidewire **134**) before selecting the rotational speed (e.g., buttons **117a-b**) and activating rotation (e.g., button on the actuator **116**). As such, the screenless user interface of the handle assembly **110** can provide a simplified and fluid hand motion for the user while also communicating effective information to the user. Optionally, the controller **150** (or **150A** in other embodiments) can be configured to prevent the electric motor **112** from driving rotation of the drive shaft **136** until: (1) the guidewire brake **118** is activated (e.g., with the guidewire brake light **119** illuminated), (2) the pump **114** is activated to drive the flush fluid (e.g., via actuation of fluid pump switch **117c** that then illuminates the button **117c**), (3) a rotation speed has been selected via speed selection switches **117a** and **117b** (e.g., with a speed indicator light thereon being activated), or a combination of all these conditions. As another example, the indicator lights associated with the selection switches **117a** and **117b**, the fluid pump switch **117c**, and the guidewire brake light **119** will alert a clinician that the rotational atherectomy system **100** should not be operated until all three systems (the motor, the pump, the guidewire brake) are activated. For example, each system may have a green light, such that three green lights indicates the clinician can proceed with the atherectomy procedure. Optionally, only the guidewire **118** needs to be actuated to allow rotation of the rotational atherectomy system **100**.

[0077] Still referring to FIG. **1**, the rotational atherectomy system **100** also includes the controller **150**, which in this embodiment which includes a processor and computer-readable memory storing control instructions thereon. The controller **150** is configured to receive input from sensors housed within the handle assembly, to receive input from the user interface on the handle assembly **110** (e.g., switches/actuators **116**, **117a-c**, and **118**), and to control the activation of the electric motor **112** and the pump **114** (responsive to inputs at the user interface switches/actuators). In this embodiment, the controller **150** is contained inside the housing **122** of the power adapter **120** so that it is reusable with subsequent handle assemblies after the first handle assembly **110** is discarded (e.g., after use with a first patient). As previously described, the cable **121** can provide data communication between the controller **150** and the components of the user interface (e.g., switches/actuators **116**, **117a-c**, and **118**), the electric motor **112**, the pump **114**, and the feedback sensors housed within the handle assembly **110**. In an alternative embodiment, the controller (including the processor and computer-readable memory storing the control instructions) can be provided in the form of controller **150A** configured to be contained within the housing **111** of the handle assembly **110** (in proximity to the electric motor **112** and the pump **114**). In both options, the handle assembly **110** can be operated by a clinician using the above-described simplified, screenless interface to perform and control the rotational atherectomy procedure (e.g., without a user interface display screen along the handle assembly or on the units connected to the handle assembly). Preferably, the controller **150** (or controller **150A**) is housed in a manner that is sealed

from fluids encountered by the handle assembly, such as saline, blood, or others.

[0078] Referring now to FIGS. 2-3, some embodiments of the distal end portion of the drive shaft **136** include an improved configuration of the abrasive burrs (and optionally the distal stability element) that provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** of the drive shaft **136** so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the abrasive elements **140a-c** are eccentrically-fixed to the torque-transmitting coil **137** of the driveshaft **136** while the distal stability element **142** (having a similar abrasive surface) is concentric with the torque-transmitting coil **137** of the drive shaft **136**. As shown in FIG. 2, the torque-transmitting coil **137** as a coil diameter A, and all of the abrasive burrs **140a-c** have the same diameter G (which is larger than the coil diameter). For example, the abrasive burrs **140a-c** can have a diameter G of 1.0 mm to 1.33 mm, preferably 1.1 mm to 1.3 mm, and 1.25 mm in the depicted embodiment here (e.g., nominal diameter of 1.25 mm before application of thin abrasive coating). And, in such examples, the coil diameter A can be 0.7 mm to 0.9 mm, and preferably 0.8 mm in this depicted embodiment. Accordingly, some embodiments described herein a relative sizing ratio between the diameter G of each abrasive element **140a-c** and the coil diameter A (e.g., a burr-to-coil diameter ratio) that is 1.7 or less. In some implementations, the burr-to-coil diameter ratio can be about 1.3-1.7 for all abrasive elements along the torque-transmitting coil of the drive shaft. In the depicted example, the burr-to-coil diameter ratio can be 1.5-1.6 for each eccentric abrasive element **140a-c** mounted to the torque-transmitting coil **137** and 1.3-1.4 for the distal stability element **142** mounted to the torque-transmitting coil **137**.

[0079] Also shown in FIG. 3, the centers of mass of the abrasive burrs **140a-c** are offset in different planes at radial spacing angle H. For example, the radial spacing angle H of 5 degrees to 87.5 degrees, preferably 20 degrees to 60 degrees, and 37.5 degrees in the depicted embodiment here. As such, the combined radial angles of the all abrasive burrs **140a-c** is less than 175 degrees (e.g., less than 120 degrees, and preferably less than 90 degrees in the depicted embodiment) along that particular length of the drive shaft carrying the abrasive burrs **140a-c**.

[0080] Still referring to FIGS. 2-3, the abrasive burrs **140a-c** can be mounted to the coil **137** so as to provide a compact end length E between the proximal-most abrasive element **140a** and the distal-most tip of the drive shaft **136**, which can be effective in treating stenotic lesions **107** in small arteries, such as those in the feet below the ankle or those in the coronary arteries. For example, the compact end length E can be 5 cm or less from the distal-most tip of the drive shaft **136**, preferably 2 cm or less from the distal-most tip of the drive shaft **136**, and about 1.9 cm (0.75 inches) in the depicted embodiment. Such an end length E can be relative compact compared to the overall length B of torque-transmitting coil **137**. For example, the overall length B can be 150 cm to 250 cm, 110 cm to 200 cm, and about 197 cm (77.5 inches) in the depicted embodiment. As such, the end length E can be less than 1% of the overall length B. In some embodiments, the ratio of the overall length B to the end length E is greater than 50:1, about 90:1 to 140:1, and about 103:1 in the depicted embodiment. Also, in the depicted embodiment, the abrasive burrs **140a-c** in the compact end length E includes a total of three discrete elements that are spaced apart from each other. In other embodiments, one, two, three, four, or five discrete abrasive elements are included in the set of abrasive elements within the compact end length E.

[0081] Additionally, within the compact end length E of the drive shaft **136**, the relative spacing of the abrasive burrs **140a-c** can advantageously affect the orbital path of the abrasive elements, the flexibility of the intermediate sections of the torque-transmitting coil (with such flexibility being useful during advancement through tortuous arterial paths), or both. For example, the distal-most abrasive burr **140c** can be spaced from the distal-most end of the shaft by an extension length C, with the next abrasive burr **140b** being spaced therefrom by a burr spacing distance D, and the proximal-most abrasive burr **140a** being spaced apart therefrom by the same burr spacing distance

D. In the depicted embodiment, the extension length C is greater than the burr spacing distance D, but they can be approximately equal in other embodiments (refer, for example, to FIGS. 21-24). For example, the extension length C can be 4 mm to 15 mm, 5 mm to 10 mm, and about 9 mm (0.35 inches) in the depicted embodiment, while the burr spacing distance D can be 2 mm to 7 mm, 4 mm to 6 mm, and about 5 mm (0.20 inches) in the depicted embodiment.

[0082] Still referring to FIGS. 2-3, some embodiments of the abrasive burrs **140a-c**, the optional distal stability element **142**, and the optional distal tip member **144** can provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** so as to advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. 1 above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery. The embodiment depicted in FIGS. 2-3 is provided in the form of a "4-French introducer" device, as described above.

[0083] Referring now to FIG. 4, the filar wind direction **138** (e.g., the winding direction of filars of the torque transmitting coil **137** traversing from the distal end toward the proximal end) and the rotational direction **139** of the torque transmitting coil **137** (e.g., the rotational direction of the drive shaft **136** urged by the handle assembly **110**) can be configured to provide a number of performance benefits during the rotational atherectomy in the pedal or coronary arteries, especially those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. For example, as shown in FIG. 4 (and also in FIGS. 2-3), the filar wind direction **138** is opposite from the rotational direction **139** of the torque transmitting coil **137**, which can provide functional benefits when advancing into tight lesions in small arteries within the foot (below the ankle) or through a tortuous path into a branch coronary artery (e.g., the left anterior descending coronary artery or the left circumflex coronary artery). In particular, during rotation of the driveshaft **136** adjacent to such a lesion, one of the abrasive elements **140a-c**, the distal stability element **142**, or the distal tip **144** may catch on the bodily material or otherwise may be momentarily restrained from rotation (even though the proximal end of the drive shaft **136** is rotated by the handle assembly **110** (FIG. 1). In such circumstances, the filar wind direction **138** is opposite from the rotational direction **139** of the torque-transmitting coil **137** as shown in FIGS. 2-3, so the torque-transmitting coil **137** will not suffer a narrowed coil diameter that causes the torque-transmitting coil **137** to clamp down on the guidewire **134** (e.g., creating reduced flexibility and potentially other concerns in the narrow space with the vessel), but instead the coil diameter will remain the same or be momentarily expanded (safely avoiding a clamping effect on the guidewire **134**).

[0084] Accordingly, the drive shaft **136** of the rotational atherectomy system **100** can have a configuration that provides safe and repeatable navigation into smaller blood vessels below the ankle or in the heart and effective orbital paths for abrading stenotic material in such smaller vessels. Such configurations can be particularly useful, for example, when implemented as a pedal loop navigable device, a coronary branch navigable device, or both.

[0085] Referring now to FIG. 5, the drive shaft **136** of the rotational atherectomy device can be equipped with the concentric tip member **144** and the distal stability element **142** in an arrangement that covers the distal-most end of the torque-transmitting coil **137**. As such, the stenotic lesion **107** (FIG. 4) is initially engaged with the concentric tip member **144** and the distal stability element **142** before engaging an exterior of the torque-transmitting coil **137**. In this embodiment, the distal tip member is welded (e.g., a butt weld) or otherwise fixed to a distal-most end of the torque-transmitting coil **137** of the drive shaft **136** so that the entirety of the distal tip member **144** is positioned axially distal of the torque-transmitting coil **137**. Also, the distal stability element **142** is a cylindrical structure in the embodiment, which is welded or otherwise fixed to the distal-most end of the torque-transmitting coil **137** so that the distal stability element **142** is positioned radially outward of the coil diameter. As such, the distal-most end of the torque-transmitting coil of the

drive shaft **136** is concealed by the distal stability element **142** and the concentric tip member **144**. In this embodiment, both the distal stability element **142** and the concentric tip member **144** comprise metallic cylindrical members that are axially aligned with a central axis of the drive shaft **136**, but they are different in size and in abrasiveness. The distal stability element **142** has an inner diameter that surrounds an exterior coil diameter of the drive shaft **136** (and thus the maximum outer diameter of the distal stability element **142** is larger than the coil diameter), and the concentric tip member **144** has an outer diameter that is substantially the same as the coil diameter. Also, the distal stability element **142** has an abrasive outer coating. For example, in some embodiments, a diamond coating (or other suitable type of abrasive coating) is disposed on the outer surface of the distal stability element **142**. The concentric tip member **144** in this embodiment has smooth exterior surface that is less abrasive than that of the distal stability element **142**. In some cases, the smooth initial surface of the concentric tip member **144** followed by the abrasive surface on the distal stability element **142** can help facilitate the initial expansion and abrasion pilot path through the stenotic lesion **107** in the targeted vessel **105**. Both the distal stability element **142** and the concentric tip member **144** may comprise a biocompatible material, such as a higher-density biocompatible material. For example, in some embodiments, each of the distal stability element **140** and the concentric tip member **144** may comprise a metallic material such as stainless steel, tungsten, molybdenum, iridium, cobalt, cadmium, and the like, and alloys thereof. Also, in this embodiment, the distal stability element **140** and the concentric tip member **144** have a fixed outer diameter. That is, the distal stability element **140** and the concentric tip member **144** are not an expandable member in the depicted embodiment

[0086] Referring now to FIGS. **6-7**, some embodiments of the rotational atherectomy system **100** can remove (partially or completely) one or more stenotic lesions in a targeted pedal artery below the ankle. For example, in the depicted implementation, the drive shaft **136** of the rotational atherectomy system **100** is navigated through the anterior tibial (AT) artery into the dorsalis pedis (DP) artery toward the pedal loop (LP). As previously described, the components along the distal end portion of the drive shaft **136** can have an orientation, relative spacing, and relative size so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In use, the system **100** includes the guidewire **134**, which is advanced into the target pedal artery below the ankle. In the example shown in FIGS. **6-7**, the guidewire **134** is navigated through the anterior tibial (AT) artery into the dorsalis pedis (DP) artery, and into the pedal loop (LP). Such a guidewire position can be useful for rotational atherectomy that targets one or more lesions in the dorsalis pedis artery (DP). It should be understood from the description here that the targeted pedal artery may additionally or alternatively include the pedal loop (LP), the anterior tibial (AT) artery below the ankle, the posterior tibial artery (PT) below the ankle, the common plantar (CP) artery, the medial plantar (MP) artery. After advancement of the guidewire **134**, the drive shaft **136** and sheath **132** (FIG. **1**) are advanced over the proximal end of the guidewire **134** such that the proximal end of the guidewire **134** passes through the entirety of the drive shaft **136** and the handle assembly **110** (FIG. **1**) to protrude rearwardly from the proximal end of the handle assembly **110**. Optionally, the drive shaft **136** and sheath **132** are advanced through a relatively small introducer sheath in a patient's leg, such as a 4-French introducer sheath (FIG. **1**).

[0087] Still referring to FIGS. **6-7**, as the distal end portion of the drive shaft **136** is navigated toward the targeted pedal artery (preferably, under medical imaging), the concentric tip member **144** (followed by the abrasive surface on the distal stability element **142**) can be used to initially form a pilot path through the stenotic lesion in the pedal artery. From there, the user can select a rotational speed setting (e.g., using user interface buttons **117a-b** on the handle assembly) so that the abrasive elements **140a-c** can abrade the stenotic material and achieve an orbital path during rotation (e.g., an orbital path that is greater in size than a maximum stationary diameter of the abrasive elements **140a-c**). The handle assembly **110** (FIG. **1**) can be used, via the actuator **116** to

translate the abrasive elements **140a-c** in reciprocating movement (distally and proximally) for multiple passes through the stenotic lesion in the pedal artery during high speed rotation of the abrasive elements **140a-c** in the orbital path.

[0088] Accordingly, the drive shaft **136** of the rotational atherectomy system **100** can have a configuration that provides safe and repeatable navigation into smaller blood vessels below the ankle or in the heart and effective orbital paths for abrading stenotic material in such smaller vessels. Such configurations can be particularly useful, for example as shown in FIGS. 2-7, when the rotational atherectomy system **100** implemented as a pedal loop navigable system. Additionally, the implementation of the drive shaft depicted in FIGS. 2-7 can be effective for the removal or reduction of stenotic lesions in larger vessels too (e.g., those in the leg above the ankle), thereby providing the user with options for efficiently treating a variety of arterial sites during a single procedure using a single driveshaft **136**.

[0089] Referring now to FIGS. 8-9, some embodiments of the distal end portion of the drive shaft **136** include an alternative configuration of the abrasive burrs that also provides a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** of the drive shaft **136** so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the components along the distal end portion of the drive shaft **136** are similar to those described in connection with FIGS. 2-3, except that the distal and proximal abrasive elements **140d** have a smaller size than the intermediate abrasive element **140b**. For example, the torque-transmitting coil **137**, the filar wind direction **138**, the rotation direction **139**, the intermediate abrasive element **140b**, the optional distal stability element **142**, and the optional distal tip member **144** have a similar configuration as previously described in connection with FIGS. 2-3. Here, as shown in FIGS. 8-9, the set of abrasive burrs along the end length of the torque-transmitting coil **137** include distal and proximal abrasive burrs **140d**, which have a size that is smaller than the intermediate abrasive burr **140b**. As shown in FIG. 8, the distal and proximal abrasive burrs **140d** have a diameter **G1** and the intermediate abrasive burr **140b** has a larger diameter **G2** (both **G1** and **G2** are larger than the coil diameter **A** described in connection with FIG. 2). For example, the distal and proximal abrasive burrs **140d** can have a diameter **G1** of 1.0 mm to 1.15 mm, and particularly 1.1 mm in the depicted embodiment here (e.g., nominal diameter of 1.1 mm before application of thin abrasive coating), and the intermediate abrasive burr **140b** can have a diameter **G2** of 1.2 mm to 1.33 mm, and particularly 1.25 mm in the depicted embodiment here (e.g., nominal diameter of 1.25 mm before application of thin abrasive coating). As previously described, in such examples, the coil diameter **A** can be 0.7 mm to 0.9 mm, and preferably 0.8 mm in this depicted embodiment. Accordingly, in the depicted example, the burr-to-coil diameter ratio can be 1.3-1.4 for each smaller abrasive burr **140d** mounted to the torque-transmitting coil **137**, and the burr-to-coil diameter ratio can be 1.5-1.7 for the larger abrasive burr **140b** mounted to the torque-transmitting coil **137**. Also shown in FIG. 9, the centers of mass of the distal abrasive burr **140d**, the intermediate abrasive burr **140b**, and the proximal abrasive burr **140d** are offset in different planes at radial spacing angle **H**. As previously described in connection with FIGS. 2-3, the radial spacing angle **H** of 5 degrees to 87.5 degrees, preferably 20 degrees to 60 degrees, and 37.5 degrees in the depicted embodiment here. As such, the combined radial angles of the all abrasive burrs **140d**, **140b**, and **140d** is less than 175 degrees (e.g., less than 120 degrees, and preferably less than 90 degrees in the depicted embodiment) along the end length of the drive shaft.

[0090] Accordingly, similar to the embodiments described above in connection with FIGS. 2-4, the embodiment depicted in FIGS. 8-9 is provided in the form of a “4-French introducer” device, as described above. Additionally, the abrasive burrs **140d**, **140b**, and **140d** shown in FIGS. 8-9, the optional distal stability element **142**, and the optional distal tip member **144** can collectively provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** so as to advantageously advance through a relatively small percutaneous access point, such as

an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. 1 above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery.

[0091] Referring now to FIGS. 10-11, additional embodiments of the distal end portion of the drive shaft 136 include an alternative configuration of the abrasive burrs that also provides a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil 137 of the drive shaft 136 so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the components along the distal end portion of the drive shaft 136 are similar to those described in connection with FIGS. 8-9, except that the distal and proximal abrasive elements 140e have a different cutout for mounting to the torque-transmitting coil 137 so that the geometric center of the spherical element is axially aligned with the axis 135 of the torque-transmitting coil 137. For example, the torque-transmitting coil 137, the filar wind direction 138, the rotation direction 139, the intermediate abrasive element 140b, the optional distal stability element 142, and the optional distal tip member 144 have a similar configuration as previously described in connection with FIGS. 2-3 and 8-9. Here, as shown in FIGS. 10-11, the distal and proximal abrasive burrs 140e have deeper mounting slot formed therein (e.g., deeper compared to the example abrasive elements 140d in FIGS. 8-9) such that, when the distal and proximal abrasive burrs 140e are affixed to the exterior of the torque-transmitting coil 137, the geometric center of each spherical element 140e is axially aligned with the axis 135 of the torque-transmitting coil 137. In other words, even though the distal and proximal abrasive burrs 140d (FIGS. 8-9) and the distal and proximal abrasive burrs 140e (FIGS. 10-11) have the same diameter size, the outer spherical surface of the distal and proximal abrasive burrs 140e (FIGS. 10-11) is closer to the central axis 135 of the torque-transmitting coil 137. Due to the mounting slot formed in each of the distal and proximal abrasive burrs 140e, the center of mass of the burr 140e is offset from the central axis while the geometric center of the burr 140e is axially aligned with the axis. Also, as shown in FIG. 11, the centers of mass of the distal abrasive burr 140e, the intermediate abrasive burr 140b, and the proximal abrasive burr 140e are offset in different planes at radial spacing angle H. As previously described in connection with FIGS. 2-3 and 8-9, the radial spacing angle H of 5 degrees to 87.5 degrees, preferably 20 degrees to 60 degrees, and 37.5 degrees in the depicted embodiment here. As such, the combined radial angles of the all abrasive burrs 140e, 140b, and 140e is less than 175 degrees (e.g., less than 120 degrees, and preferably less than 90 degrees in the depicted embodiment) along the end length of the drive shaft.

[0092] Here again, similar to the embodiments described above in connection with FIGS. 2-4 and 8-9, the embodiment depicted in FIGS. 10-11 is provided in the form of a "4-French introducer" device, as described above. Additionally, the abrasive burrs 140e, 140b, and 140e shown in FIGS. 10-11, the optional distal stability element 142, and the optional distal tip member 144 can collectively provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil 137 so as to advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. 1 above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery.

[0093] Referring now to FIGS. 12-13, additional embodiments of the distal end portion of the drive shaft 136 include an alternative configuration of the abrasive burrs that also provides a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil 137 of the drive shaft 136 so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the components along the distal end portion of the drive shaft 136 are similar to those described in connection with FIGS. 8-9, except that one of the distal and proximal abrasive elements (the distal abrasive element 140f in this example) has a center of mass that is

offset in the same plane as the intermediate abrasive element **140b**. For example, the torque-transmitting coil **137**, the filar wind direction **138**, the rotation direction **139**, the intermediate abrasive element **140b**, the optional distal stability element **142**, and the optional distal tip member **144** have a similar configuration as previously described in connection with FIGS. 2-3 and 8-9. And, similar to the embodiment shown in FIGS. 8-9, the distal abrasive burr **140e** and proximal abrasive burr **140d** have the same diameter size (e.g., diameter size G1 described above in FIG. 8), which is smaller than the intermediate abrasive burr **140b** (e.g., diameter size G2 described above in FIG. 8). However, as shown in FIGS. 12-13, the abrasive burrs are offset only in two longitudinal planes. For example, only one of the abrasive burrs (element **140d** in this example) has a center of mass that is offset in a different plane at radial spacing angle H, while the remaining abrasive burrs **140f** and **140b** have centers of mass that are offset in the same longitudinal plane. As previously described in connection with FIGS. 2-3 and 8-9, the radial spacing angle H of 5 degrees to 87.5 degrees, preferably 20 degrees to 60 degrees, and 37.5 degrees in the depicted embodiment here. As such, the combined radial angles of the all abrasive burrs **140e**, **140b**, and **140e** is less than 175 degrees (e.g., less than 120 degrees, and preferably less than 90 degrees in the depicted embodiment) along the end length of the drive shaft.

[0094] Accordingly, similar to the embodiments described above in connection with FIGS. 2-4 and 8-11, the embodiment depicted in FIGS. 12-13 is provided in the form of a “4-French introducer” device, as described above. Additionally, the abrasive burrs **140f**, **140b**, and **140d** shown in FIGS. 12-13, the optional distal stability element **142**, and the optional distal tip member **144** can collectively provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** so as to advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. 1 above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery.

[0095] Referring now to FIGS. 14-15, further embodiments of the distal end portion of the drive shaft **136** include an alternative configuration of the abrasive burrs that also provides a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** of the drive shaft **136** so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the components along the distal end portion of the drive shaft **136** are similar to those described in connection with FIGS. 8-9, except that all of the distal and proximal abrasive elements **140f** and the intermediate abrasive element have centers of mass offset in the same plane. For example, the torque-transmitting coil **137**, the filar wind direction **138**, the rotation direction **139**, the intermediate abrasive element **140b**, the optional distal stability element **142**, and the optional distal tip member **144** have a similar configuration as previously described in connection with FIGS. 2-3 and 8-9. And, similar to the embodiment shown in FIGS. 8-9, the distal abrasive burr **140f** and proximal abrasive burr **140f** have the same diameter size (e.g., diameter size G1 described above in FIG. 8), which is smaller than the intermediate abrasive burr **140b** (e.g., diameter size G2 described above in FIG. 8). However, as shown in FIGS. 14-15, the abrasive burrs are offset only in a single longitudinal plane.

[0096] Here again, similar to the embodiments described above in connection with FIGS. 2-4 and 8-13, the embodiment depicted in FIGS. 14-15 is provided in the form of a “4-French introducer” device, as described above. Additionally, the abrasive burrs **140f**, **140b**, and **140f** shown in FIGS. 14-15, the optional distal stability element **142**, and the optional distal tip member **144** can collectively provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** so as to advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. 1 above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery.

[0097] Referring now to FIG. 16, some embodiments of the rotational atherectomy system **100** can advance through the tortuous path into a branch coronary artery (e.g., the left anterior descending coronary artery or the left circumflex coronary artery) of a heart to remove (partially or completely) one or more stenotic lesions in a targeted branch coronary artery. For example, in the depicted implementation, the drive shaft **136** of the rotational atherectomy system **100** is navigated through the aorta into the left main artery (LMA) toward the left anterior descending artery (LAD) of the heart. In the depicted embodiment, the distal end portion of the drive shaft **136** includes the configuration depicted in FIGS. 14-15, but it should be understood from the description herein that other embodiments (e.g., FIGS. 2-3, 8-13, and 17-24) can also be implemented in the targeted branch coronary artery.

[0098] As previously described, the components along the distal end portion of the drive shaft **136** can have an orientation, relative spacing, and relative size so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In use, the system **100** includes the guidewire **134**, which is advanced into the target branch coronary artery of the heart. In the example shown in FIG. 16, the guidewire **134** is navigated through the aorta into the left main artery (LMA) toward the left anterior descending artery (LAD) of the heart. Such a guidewire position can be useful for rotational atherectomy that targets one or more lesions in the left anterior descending artery (LAD). It should be understood from the description here that the targeted branch coronary artery may alternatively include the left circumflex artery (LCX) or other branch artery along the heart. After advancement of the guidewire **134**, the drive shaft **136** and sheath **132** (FIG. 1) are advanced over the proximal end of the guidewire **134** such that the proximal end of the guidewire **134** passes through the entirety of the drive shaft **136** and the handle assembly **110** (FIG. 1) to protrude rearwardly from the proximal end of the handle assembly **110**. Optionally, the drive shaft **136** and sheath **132** are advanced through a relatively small introducer sheath in a patient's leg, such as a 4-French introducer sheath (FIG. 1).

[0099] Still referring to FIG. 16, as the distal end portion of the drive shaft **136** is navigated toward the targeted branch coronary artery (preferably, under medical imaging), the concentric tip member **144** (followed by the abrasive surface on the distal stability element **142**) can be used to initially form a pilot path through the stenotic lesion in the branch coronary artery. From there, the user can select a rotational speed setting (e.g., using user interface buttons **117a-b** on the handle assembly) so that the abrasive elements **140f**, **140b**, and **140f** can abrade the stenotic material and achieve an orbital path during rotation (e.g., an orbital path that is greater in size than a maximum stationary diameter of the larger abrasive element **140b**). The handle assembly **110** (FIG. 1) can be used, via the actuator **116** to translate the abrasive elements **140f**, **140b**, and **140f** in reciprocating movement (distally and proximally) for multiple passes through the stenotic lesion in the pedal artery during high speed rotation of the abrasive elements **140f**, **140b**, and **140f** in the orbital path.

[0100] Accordingly, the drive shaft **136** of the rotational atherectomy system **100** can have a configuration that provides safe and repeatable navigation into smaller blood vessels below the ankle or in the heart and effective orbital paths for abrading stenotic material in such smaller vessels. Such configurations can be particularly useful, for example as shown in FIGS. 2-7, when the rotational atherectomy system **100** implemented as a pedal loop navigable system. Additionally, the implementation of the drive shaft depicted in FIGS. 2-7 can be effective for the removal or reduction of stenotic lesions in larger vessels too (e.g., those in the leg above the ankle), thereby providing the user with options for efficiently treating a variety of arterial sites during a single procedure using a single driveshaft **136**.

[0101] Referring now to FIGS. 17-18, additional embodiments of the distal end portion of the drive shaft **136** include an alternative configuration of the abrasive burrs that also provides a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** of the drive shaft **136** so as to achieve an efficient access path to such small arteries, including those where the

vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the components along the distal end portion of the drive shaft **136** are similar to those described in connection with FIGS. **8-9**, except that the distal and proximal abrasive elements **140g** have a cylindrical shape and are mounted in axial alignment with the axis **135** of the torque-transmitting coil **137**. For example, the torque-transmitting coil **137**, the filar wind direction **138**, the rotation direction **139**, the intermediate abrasive element **140b**, the optional distal stability element **142**, and the optional distal tip member **144** have a similar configuration as previously described in connection with FIGS. **2-3** and **8-9**. Here, as shown in FIGS. **17-18**, the distal and proximal abrasive burrs **140g** have a tubular shape that is similar to that of the distal stability element **142** (refer also to FIG. **5** for details of the tubular structure) such that, when the distal and proximal abrasive burrs **140g** are affixed to the exterior of the torque-transmitting coil **137**, the center of mass and the geometric center of each cylindrical element **140g** are axially aligned with the central axis **135** of the torque-transmitting coil **137**. Also, the distal and proximal abrasive burrs **140g** (FIGS. **17-18**) have the same diameter size (e.g., diameter size G1 described above in FIG. **8**), which is smaller than the intermediate abrasive burr **140b** (e.g., diameter size G2 described above in FIG. **8**). However, as shown in FIGS. **17-18**, only one of the abrasive burrs (element **140b** in this example) has a center of mass offset from the central axis **135**. Similar to the embodiments described above in connection with FIGS. **2-4** and **8-15**, the embodiment depicted in FIGS. **17-18** is provided in the form of a “4-French introducer” device, as described above. Additionally, the abrasive burrs **140g**, **140b**, and **140g** shown in FIGS. **17-18**, the optional distal stability element **142**, and the optional distal tip member **144** can collectively provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** so as to advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. **1** above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery.

[0102] Referring now to FIGS. **19-20**, additional embodiments of the distal end portion of the drive shaft **136** include an alternative configuration of the abrasive burrs that also provides a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** of the drive shaft **136** so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the components along the distal end portion of the drive shaft **136** are similar to those described in connection with FIGS. **17-18**, except that the distal and proximal abrasive elements **140h** have a cylindrical shape with a shorter axial length than the cylindrical elements **140g** depicted in FIGS. **17-18**. For example, the torque-transmitting coil **137**, the filar wind direction **138**, the rotation direction **139**, the intermediate abrasive element **140b**, the optional distal stability element **142**, and the optional distal tip member **144** have a similar configuration as previously described in connection with FIGS. **2-3** and **8-9**. Here, as shown in FIGS. **19-20**, the distal and proximal abrasive burrs **140h** have a tubular shape with a diameter similar to that of the distal stability element **142** (refer also to FIG. **5** for details of the tubular structure) but with an axial length that shorter than that of the distal stability element **142** (e.g., less than half of the axial length of the distal stability element **142** in this embodiment). For example, the axial length of each of the distal and proximal abrasive burrs **140h** is 0.6 mm to 0.8 mm, and particularly 0.76 mm in the embodiment (which is also less than the coil diameter in this embodiment). In this example, the axial length of distal abrasive element **142** is 1.75 mm to 2 mm, and particularly 1.9 mm in the embodiment (which is greater than the coil diameter in this embodiment). As previously described, the distal and proximal abrasive burrs **140h** are affixed to the exterior of the torque-transmitting coil **137** so that the center of mass and the geometric center of each cylindrical element **140h** are axially aligned with the central axis **135** of the torque-transmitting coil **137**. Also, the distal and proximal abrasive burrs **140h** (FIGS. **19-20**) have the same diameter size (e.g., diameter size G1

described above in FIG. 8), which is smaller than the intermediate abrasive burr **140b** (e.g., diameter size G2 described above in FIG. 8). As shown in FIGS. 19-20, only one of the abrasive burrs (element **140b** in this example) has a center of mass offset from the central axis **135**. Similar to the embodiments described above in connection with FIGS. 2-4, 8-15 and 17-18, the embodiment depicted in FIGS. 17-18 is provided in the form of a “4-French introducer” device, as described above. Additionally, the abrasive burrs **140g**, **140b**, and **140g** shown in FIGS. 19-20, the optional distal stability element **142**, and the optional distal tip member **144** can collectively provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** so as to advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. 1 above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery.

[0103] Referring now to FIGS. 21-22, additional embodiments of the distal end portion of the drive shaft **136** include an alternative configuration of the abrasive burrs that also provides a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** of the drive shaft **136** so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the components along the distal end portion of the drive shaft **136** are similar to those described in connection with FIGS. 2-3, except that (i) those components are mounted within a more compact end length E' (shorter than end length E shown in FIG. 2), (ii) the distal and proximal abrasive elements **140i** have a smaller size than the intermediate abrasive element **140b**, and (iii) the distal stability element **142'** has a smaller size (smaller than the distal stability element **142** shown in FIG. 2). For example, the torque-transmitting coil **137**, the filar wind direction **138**, the rotation direction **139**, the intermediate abrasive element **140b**, and the optional distal tip member **144** have a similar configuration as previously described in connection with FIGS. 2-3. Here, as shown in FIGS. 21-22, the abrasive burrs **140i**, **140b**, and **140i** can be mounted to the coil **137** so as to provide a more compact end length E' between the proximal-most abrasive element **140i** and the distal-most tip of the drive shaft **136**, which can be effective in treating stenotic lesions **107** in small arteries, such as those in the feet below the ankle or those in the coronary arteries. For example, the more compact end length E can be less than 1.8 cm from the distal-most tip of the drive shaft **136**, preferably 1.3 cm to 1.6 cm, and about 1.5 cm (0.60 inches) in the depicted embodiment. Such an end length E can be relative compact compared to the overall length of torque-transmitting coil **137** (e.g., refer to length B in FIG. 2, which can be 150 cm to 250 cm, and about 197 cm (77.5 inches) in this embodiment). As such, the end length E' can be less than 0.8% of the overall length B. In some embodiments, the ratio of the overall length B to the end length E is about 120:1 to 140:1, and about 129:1 in the depicted embodiment. Also, in the depicted embodiment, the abrasive burrs **140i**, **140b**, and **140i** in the more compact end length E' includes a total of three discrete elements that are spaced apart from each other. In other embodiments, one, two, three, four, or five discrete abrasive elements are included in the set of abrasive elements within the more compact end length E'. Additionally, within the more compact end length E' of the drive shaft **136**, the distal-most abrasive burr **140i** can be spaced from the distal-most end of the shaft by an extension length C', with the intermediate abrasive burr **140b** being spaced therefrom by a burr spacing distance D', and the proximal-most abrasive burr **140i** being spaced apart therefrom by the same burr spacing distance D'. In the depicted embodiment, the extension length C' is the same as the burr spacing distance D'. For example, the extension length C' can be 4 mm to 6 mm, and about 5 mm (0.20 inches) in the depicted embodiment, while the burr spacing distance D' can likewise be 4 mm to 6 mm, and about 5 mm (0.20 inches) in the depicted embodiment.

[0104] Still referring to FIGS. 21-22, the set of abrasive burrs along the end length E' of the torque-transmitting coil **137** include the distal and proximal abrasive burrs **140i**, which have a size that is smaller than the intermediate abrasive burr **140b**. As shown in FIG. 21, the distal and proximal

abrasive burrs **140d** have a diameter **G3** and the intermediate abrasive burr **140b** has a larger diameter **G2** (both **G3** and **G2** are larger than the coil diameter **A** described in connection with FIG. 2). For example, the distal and proximal abrasive burrs **140i** can have a diameter **G3** of 0.9 mm to 1.1 mm, and particularly 1.0 mm in the depicted embodiment here (e.g., nominal diameter of 1.0 mm before application of thin abrasive coating), and the intermediate abrasive burr **140b** can have a diameter **G2** of 1.2 mm to 1.33 mm, and particularly 1.25 mm in the depicted embodiment here (e.g., nominal diameter of 1.25 mm before application of thin abrasive coating). As previously described, in such examples, the coil diameter **A** can be 0.7 mm to 0.9 mm, and preferably 0.8 mm in this depicted embodiment. Accordingly, in the depicted example, the burr-to-coil diameter ratio can be 1.2-1.3 for each smaller abrasive burr **140i** mounted to the torque-transmitting coil **137**, and the burr-to-coil diameter ratio can be 1.5-1.7 for the larger abrasive burr **140b** mounted to the torque-transmitting coil **137**. Also shown in FIG. 22, the centers of mass of the distal abrasive burr **140i**, the intermediate abrasive burr **140b**, and the proximal abrasive burr **140i** are offset in different planes at radial spacing angle **H**. As previously described in connection with FIGS. 2-3, the radial spacing angle **H** of 5 degrees to 87.5 degrees, preferably 20 degrees to 60 degrees, and 37.5 degrees in the depicted embodiment here. As such, the combined radial angles of the all abrasive burrs **140i**, **140b**, and **140i** is less than 175 degrees (e.g., less than 120 degrees, and preferably less than 90 degrees in the depicted embodiment) along the end length **E'** of the drive shaft.

[0105] Still referring to FIGS. 21-22, the some embodiments of the drive shaft **136** can include distal stability element **142'** having a shorter axial length. For example, the distal stability element **142'** can have an axial length that smaller than the coil diameter of the torque-transmitting coil **137** (refer to coil diameter **A** described in connection with FIG. 2). In this embodiment, the axial length of the distal stability element **142'** is 0.6 mm to 0.8 mm, and particularly 0.76 mm in the embodiment. Here, the distal stability element **142'** has a cylindrical shape and can have an outer diameter that is greater than the distal and proximal abrasive burrs **140i** (e.g., diameter size **G3** described above), which is smaller than the intermediate abrasive burr **140b** (e.g., diameter size **G2** described above). Similar to the embodiments described above in connection with FIGS. 2-4, 8-15 and 17-20, the embodiment depicted in FIGS. 21-22 is provided in the form of a "4-French introducer" device, as described above. Additionally, the abrasive burrs **140i**, **140b**, and **140i** shown in FIGS. 21-22, the optional distal stability element **142'**, and the optional distal tip member **144** can collectively provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** so as to advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. 1 above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery.

[0106] Referring now to FIGS. 23-24, additional embodiments of the distal end portion of the drive shaft **136** include an alternative configuration of the abrasive burrs that also provides a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** of the drive shaft **136** so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the components along the distal end portion of the drive shaft **136** are similar to those described in connection with FIGS. 21-22, except that all of the distal abrasive element **140c**, the intermediate abrasive element **140b**, and the proximal abrasive element **140a** have the same diameter size (refer to diameter **G** described above in connection with FIG. 2). For example, the torque-transmitting coil **137**, the filar wind direction **138**, the rotation direction **139**, the intermediate abrasive element **140b**, the optional distal stability element **142'**, and the optional distal tip member **144** have a similar configuration as previously described in connection with FIGS. 21-22. And, similar to the embodiment shown in FIGS. 2-3, the distal abrasive burr **140c** and the proximal abrasive burr **140a** have the same diameter size as the intermediate abrasive burr

140b. However, as shown in FIGS. **23-24**, the abrasive burrs **140a-c** are mounted in the more compact end length **E'** (described above in FIGS. **21-22**). Also shown in FIG. **24**, the centers of mass of the distal abrasive burr **140c**, the intermediate abrasive burr **140b**, and the proximal abrasive burr **140a** are offset in different planes at radial spacing angle **H**. As previously described in connection with FIGS. **2-3**, the radial spacing angle **H** of 5 degrees to 87.5 degrees, preferably 20 degrees to 60 degrees, and 37.5 degrees in the depicted embodiment here. As such, the combined radial angles of the all abrasive burrs **140a-c** is less than 175 degrees (e.g., less than 120 degrees, and preferably less than 90 degrees in the depicted embodiment) along the more compact end length **E'** of the drive shaft **136**.

[0107] Accordingly, similar to the embodiments described above in connection with FIGS. **2-4**, **8-15** and **17-22**, the embodiment depicted in FIGS. **23-24** is provided in the form of a “4-French introducer” device, as described above. Additionally, the abrasive burrs **140a-c** shown in FIGS. **23-24**, the optional distal stability element **142'**, and the optional distal tip member **144** can collectively provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** so as to advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. **1** above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery.

[0108] Referring now to FIGS. **25-26**, additional embodiments of the distal end portion of the drive shaft **136** include an alternative configuration of the abrasive burrs that also provides a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** of the drive shaft **136** so as to achieve an efficient access path to such small arteries, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. In the depicted embodiment, the components along the distal end portion of the drive shaft **136** are similar to those described in connection with FIGS. **19-20**, except that the distal and proximal abrasive elements **140j** coaxially mounted to the torque transmitting coil **137** have a spherical shape (rather than the cylindrical shape of element **140h** in FIG. **19**) and the optional distal stability element **142** is removed (such that the abrasive elements **140j** and **140b** are positioned closer to the distal-most end as compared to FIG. **19**). For example, the torque-transmitting coil **137**, the filar wind direction **138**, the rotation direction **139**, the intermediate abrasive element **140b**, and the optional distal tip member **144** have a similar configuration as previously described in connection with FIGS. **2-3** and **8-9**. Here, as shown in FIGS. **25-26**, the distal and proximal abrasive burrs **140j** have a spherical shape that is mounted coaxially to the torque transmitting coil **137**, and the intermediate abrasive burr **140b** has a center of mass that is offset from the central axis **135**. Much like the concentric abrasive burrs **140h** depicted in FIG. **19**, the distal and proximal abrasive burrs **140j** in this embodiment are affixed to the exterior of the torque-transmitting coil **137** so that the center of mass and the geometric center of each spherical element **140j** are axially aligned with the central axis **135** of the torque-transmitting coil **137**. Also, the distal and proximal abrasive burrs **140j** (FIGS. **25-26**) have the same diameter size (e.g., diameter size **G1** described above in FIG. **8**), which is smaller than the intermediate abrasive burr **140b** (e.g., diameter size **G2** described above in FIG. **8**). In some embodiments, the distal and proximal abrasive burrs **140j** have the same diameter size of 1.0 mm to 1.15 mm, and particularly 1.1 mm in the depicted embodiment here (e.g., nominal diameter of 1.1 mm before application of thin abrasive coating), and the intermediate abrasive burr **140b** can have a diameter of 1.2 mm to 1.33 mm, and particularly 1.25 mm in the depicted embodiment here (e.g., nominal diameter of 1.25 mm before application of thin abrasive coating). As shown in the example in FIGS. **25-26**, only one of the abrasive burrs **140j**, **140b**, and **140j** mounted to the shaft **137** (element **140b** in this example) has a center of mass offset from the central axis **135**. Additionally, the distal-most abrasive burr **140j** can be spaced from the distal-most face of the shaft **136** by a tip length **C''**, with the intermediate abrasive burr **140b** being spaced therefrom by a burr spacing distance **D''**, and the proximal-most abrasive burr **140j** being spaced

apart therefrom by the same burr spacing distance D". In the depicted embodiment, the tip length C" is a different length than the burr spacing distance D". For example, the tip length C" can be 0.6 mm to 1.1 mm, and about 1 mm (preferably about 0.04 inches) in the depicted embodiment, while the burr spacing distance D" can be 4 mm to 6 mm, and about 5 mm (preferably about 0.20 inches) in the depicted embodiment. Accordingly, in such embodiments, all of the abrasive burrs **140j**, **140b**, and **140j** in the series are positioned significantly close to the distal-most tip of the drive shaft **136**. For example, in this embodiment, the proximal-most abrasive burr **140j** can be coaxially mounted to the torque-transmitting coil at a position that is less than one inch from the distal-most tip of the drive shaft **136**, and preferably no greater than 0.5 inches from the distal-most end of the torque-transmitting coil.

[0109] Similar to the embodiments described above in connection with FIGS. 2-4, 8-15 and 17-18, the embodiment depicted in FIGS. 25-26 is provided in the form of a "4-French introducer" device, as described above. Additionally, the abrasive burrs **140j**, **140b**, and **140j** shown in FIGS. 25-26, and the optional distal tip member **144** can collectively provide a relative orientation, relative spacing, and relative sizing along the torque-transmitting coil **137** so as to advantageously advance through a relatively small percutaneous access point, such as an access point in a patient's upper leg having a small introducer sheath (e.g., refer to FIG. 1 above) for subsequent advancement into the patient's foot or heart to remove (fully or partially) stenotic material from a targeted artery.

[0110] Referring to FIG. 27, in some embodiments a rotational atherectomy system **200** for removing (partially or completely) a stenotic lesion **207** from a targeted blood vessel **205** can include an actuator handle assembly **210** that controls movement of an elongate flexible drive shaft assembly **230**. The drive shaft assembly **230** includes a flexible drive shaft **236**, and a distal end portion of the driveshaft **236** includes one or more abrasive elements **240a-c** configured to abrade the stenotic lesion **207** in the targeted vessel **205**. In some embodiments, the rotational atherectomy system **200** shares features with the rotational atherectomy system **100** (e.g., including power source **128**, power adapter **120**, and a fluid source **125**).

[0111] The abrasive elements **240a-c** can have a selected configuration and relative sizing along the distal end portion of the drive shaft **236** so as to improve navigation into smaller blood vessels below the ankle or in the heart while also achieving an effective orbital path for abrading the stenotic material **207** in those vessels. (In the depicted example, the targeted blood vessel **205** has a vessel diameter VD of 3 mm or less, and about 2.5 mm as shown, while the initial path through the stenotic lesion is significantly smaller than that.) Optionally, the abrasive elements **240** and drive shaft **236** can have a selected configuration and relative sizing (e.g., FIGS. 2-3, 8-15, 17-26) that advantageously provides advancement through a small percutaneous introducer **208** (e.g., sized to slidably receive instruments of 4-French diameter or smaller) at a percutaneous opening **209** in a patient's leg, and can further navigate through a stenotic lesion **207** in a small artery **205**, such as those below the ankle (e.g., in the feet) or coronary arteries (e.g., in the left anterior descending coronary artery or the left circumflex coronary artery) prior to sweeping a larger orbital path (during rotation of the driveshaft **236**) for abrading the stenotic material **207**. For example, in the depicted implementation shown in FIG. 27, the drive shaft **236** and abrasive elements **240a-c** have the configuration described above in connection with FIG. 26-25 (e.g., drive shaft **136** and abrasive elements **140j**, **140b**, and **140j**). It should be understood from the description herein that other configurations of the abrasive elements and features along the distal end of the drive shaft (such as those options depicted in FIGS. 2-3, 8-15, and 17-24) can be implemented in the system **200** of FIG. 27.

[0112] The system **200** can also include the power adapter **120** and the fluid source **125** (e.g., a saline bag) connectable to the actuator handle assembly **210**, and the actuator handle assembly **210** can house therein an electric motor **212** (configured to drive rotation of the driveshaft **236**) and fluid pump **214** (configured to urge a fluid such as saline toward the distal end portion of the driveshaft **236**). As described in more detail below, the controller **150** for activating the electric

motor **212** and the pump **214** (responsive to inputs at the user interface buttons **216** and **217a-c** of the handle assembly **210** in addition to the position of the user interface button **216** along the actuator slot **213**) can be contained inside a housing **122** of the power adapter **120** so that it is reusable with subsequent handle assemblies after the first handle assembly **210** is discarded (a single-use handle assembly). Alternatively, a controller **250A** for operating the electric motor **212** and the pump **214** can be contained within the housing in the handle assembly **210** (in proximity to the electric motor **212** and the pump **214**), and the entire handle assembly **210** can be discarded after a single use with a patient. In both options, the handle assembly **210** can be operated by a clinician using a simplified, screenless interface to perform and control the rotational atherectomy procedure (e.g., without a graphic display screen along the handle assembly or on a separate unit connected to the handle assembly).

[0113] Still referring to FIG. **27**, the elongate flexible drive shaft assembly **230** includes a sheath **232** that extends over the flexible drive shaft **236** such that the abrasive elements **240a-c** on the distal end portion of the drive shaft **236** are housed within a distal-most end of the sheath **232** in a proximal position (e.g., retracted position shown in FIGS. **28** and **29**). In this position, the actuator **216** is positioned at the rearmost proximal end of the slot **213**. Responsive to the clinician moving (translating) the actuator **216** distally along the slot **213** for a first translation distance **203** (distally from the rearmost proximal end of the slot **213**), the abrasive elements **240a-c** are extended distally from the distal most end of the sheath **232** (e.g., FIGS. **30-31**). From there, the clinician can translate the actuator **216** distally along the slot **213** for a second translation distance **204** (e.g., FIGS. **32-33**), can optionally reciprocate the actuator **216** between the extended positions in FIGS. **31** and **33** during rotation of the drive shaft **236**). Additionally, responsive to the clinician moving (translating) the actuator **216** proximally along the slot **213** at the rearmost proximal end of the slot **213** (FIG. **28**), the abrasive elements **240a-c** are retracted and housed within the distal most end of the sheath **232** (e.g., FIGS. **28-29**). A proximal end of the sheath **232** is fixed to a distal end of the handle assembly **210**. The flexible drive shaft **236** is slidably and rotatably disposed within a lumen of the sheath **232**. The flexible drive shaft **236** defines a longitudinal lumen in which a guidewire **234** is slidably disposed. The guidewire **234** can extend through the handle assembly **210**, the sheath **232**, and the drive shaft **236** such that a proximal end of the guidewire **234** protrudes proximally from a rear port of a guidewire brake **218** at a proximal end of the handle assembly **210** while a distal end of the guidewire **234** extends distally of a distal-most end of the drive shaft **236**. In this embodiment, the flexible drive shaft **236** includes a torque-transmitting coil of one or more helically wound filars that defines the longitudinal lumen along a central longitudinal axis. The drive **236** shaft is configured to rotate about the longitudinal axis while the sheath **232** remains generally stationary. Hence, during a rotational atherectomy procedure, the sheath **232** and the guidewire **234** are generally stationary while the flexible drive shaft **236** is controllably moved (e.g., rotating about the longitudinal axis and periodically longitudinally translating proximally and/or distally). The distal end of the sheath **232** includes a marker **233** such as a radiopaque marker that can be used during an atherectomy procedure to image the position of the distal end of the sheath **232**.

[0114] In the depicted embodiment, the distal end portion of the driveshaft **236** (that can be selectively retracted into the sheath **232** and extended from the sheath **232**) includes one or more abrasive elements **240a-c**, and a (optional) concentric tip member **244** (refer to concentric tip member **144** in FIGS. **25-26**). In the depicted embodiment, the one or more abrasive elements includes a set of three eccentric abrasive elements **240a-c** that are fixedly mounted to an exterior of the torque-transmitting coil of the driveshaft **236** such that the distal and proximal abrasive burrs **240h** have a spherical shape that is mounted coaxially to the torque transmitting coil **237**, and the intermediate abrasive burr **240b** has a center of mass that is offset from the central axis **235** (e.g., as in FIGS. **25-26**). As such, the center of mass of each of the distal and proximal abrasive burrs **240h** are aligned with the central axis of the drive shaft **236** while the center of mass of the intermediate

abrasive burr **240b** is offset from the central axis of the drive shaft **236**. The concentric tip member **244** (refer to concentric tip member **144** in FIG. 6) is affixed to, and extends distally from, the terminal distal-most end of the torque-transmitting coil. As described in more detail above, the concentric tip member **244** has a smoother surface than the abrasive surfaces of the abrasive elements **240a-c**, and the concentric tip member **244** can be configured to provide initial penetration (and, optionally, dilation) through the stenotic lesion **207** in the targeted vessel **205**. [0115] Still referring to FIG. 27, as the drive shaft **236** is rotated about its longitudinal axis, the eccentric abrasive elements **240a-c** (and the portion of the drive shaft **236** to which the one or more abrasive elements **240a-c** are affixed) will be urged in an orbit path relative to the central axis of the drive shaft **236** (also as described above with respect to drive shaft **136**, for example, in connection with FIG. 5). In general, faster speeds (rpm) of rotation of the drive shaft **236** will result in larger diameters of the orbit (within the limits of the vessel diameter). The orbiting one or more abrasive elements **240a-c** will contact the stenotic lesion **207** to abrade the lesion to a reduced size with each traversal path through the lesion **207** (i.e., small particles of the lesion will be abraded from the lesion). As described further below, contemporaneous with the rotation of the drive shaft **236**, the drive shaft **236** can be translated back and forth (distally and proximally) along the longitudinal axis of the drive shaft **236**. Hence, the stenotic lesion **207** can be abraded radially and longitudinally by virtue of the simultaneous translation and orbital rotation of the abrasive elements **240a-c**.

[0116] Additionally, the torque-transmitting coil of the flexible drive shaft **236** is laterally flexible (similar to flexible drive shaft **136**) so that the drive shaft **236** can readily advance through a tortuous arterial path (e.g., in the pedal loop or in a coronary branch artery), and so that a portion of the drive shaft **236** at, and adjacent to, the one or more abrasive elements **240** can laterally deflect when acted on by the centrifugal forces resulting from the rotation of the one or more eccentric abrasive elements **240**. In the depicted embodiment, the drive shaft **236** comprises one or more helically wound wires (or filars) that provides a uniform coil diameter that is less than the diameters of all of the abrasive elements **240a-c**. As described in more detail above, this relative sizing is referred to as the burr-to-coil diameter ratio, and the burr-to-coil diameter ratio can be about 1.3-1.7 for all abrasive burrs (elements **240a-c**) along the torque-transmitting coil of the drive shaft. As such, the torque-transmitting coil of the flexible drive shaft **236** can achieve both sufficient lateral flexibility during navigation through a tortuous path (e.g., in a patient's foot or heart) and sufficient longitudinal rigidity to be pushed through a stenotic lesion (while transmitting torque to rotate the abrasive elements **240a-c**) in a small artery. In some embodiments, the flexible drive shaft **236** shares features with the flexible drive shaft **136**.

[0117] Still referring to FIG. 27, the one or more abrasive elements **240a-c** (each of which may also be referred to as an abrasive burr) can comprise a biocompatible material that is coated with an abrasive media such as diamond grit, diamond particles, silicon carbide, and the like. In the depicted embodiment, the abrasive elements **240a-c** includes a total of three discrete abrasive spheres/cylinders that are spaced apart from each other to facilitate both navigation to, and orbital abrading within, a targeted small artery, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. As previously described, in the depicted implementation shown in FIG. 27, the drive shaft **236** and abrasive elements **240a-c** have the configuration described above in connection with FIG. 26-25 (e.g., drive shaft **136** and abrasive elements **140j**, **140b**, and **140j**). As such, the distal and proximal abrasive burrs **240h** (as in **140j** of FIGS. 25-26) have the same diameter size (e.g., diameter size G1 described above in FIG. 8), which is smaller than the intermediate abrasive burr **240b** (e.g., diameter size G2 described above in FIG. 8). In some embodiments, the distal and proximal abrasive burrs **240h** have the same diameter size of 1.0 mm to 1.15 mm, and particularly 1.1 mm in the depicted embodiment here (e.g., nominal diameter of 1.1 mm before application of thin abrasive coating), and the intermediate abrasive burr **240b** can have a diameter of 1.2 mm to 1.33 mm, and particularly 1.25 mm in the

depicted embodiment here (e.g., nominal diameter of 1.25 mm before application of thin abrasive coating). The abrasive burrs **240a-c** may be mounted to the exterior of the torque-transmitting coil of the drive shaft **236** using a biocompatible adhesive, high temperature solder, welding, press fitting, and the like. Alternatively, the one or more abrasive elements **240a-c** can be integrally formed as a unitary structure with the filars of the drive shaft **236** (e.g., using filars that are wound in a different pattern to create an axially offset structure, or the like).

[0118] Still referring to FIG. 27, the rotational atherectomy system **200** also includes the actuator handle assembly **210**. The actuator handle assembly **200** can share features with the actuator handle assembly **110**. For example, the actuator handle assembly **210** includes a housing **211** and an internal carriage assembly **245** (FIGS. 34-35) that translates along an actuator slot **213**. In some embodiments, a user can grasp the actuator **216** to urge movement along the actuator slot **213**, which causes the internal carriage assembly **245** to slidably translate along the longitudinal axis of the handle assembly **210**, as indicated by the arrow **215**. In some embodiments the carriage assembly can be translated, without limitation, about 8 cm to about 12 cm, or about 6 cm to about 10 cm, or about 4 cm to about 8 cm, or about 6 cm to about 14 cm. As the carriage assembly is translated in relation to the housing **211**, the drive shaft **236** translates in relation to the sheath **232** in a corresponding manner. As previously described, the actuator **216** of the handle assembly **210** includes a rotational power button that can activate the electrical motor **212** (carried by the internal carriage assembly) to drive the rotation of the drive shaft **236**. As such, by reciprocating actuator **216** and engaging the power button, the user can reciprocate the distal end portion of the drive shaft **236** in distal and proximal directions relative to the stenotic lesion **207** while the abrasive elements **240a-c** are rotated in an orbital path and positioned distally of a distal most end of the sheath **232**. Additionally, the user can retract and extend the distal portion of the drive shaft **236** (including the abrasive elements **240a-c**) relative to the distal end of the sheath **232** to retract the abrasive elements **240a-c** into the sheath **232** and to extend the abrasive elements **240a-c** from the distal end of the sheath **232** to expose the abrasive elements **240a-c**.

[0119] The handle assembly **210** has the cable connection **121** with the power adapter **120** (configured to receive electrical power from a power source **128** such as a wall plug) and fluid line connection **126** with a saline source **125**. The cable **121** can communicate both power and data (e.g., when the controller **150** is housed within the power adapter housing **122**), or alternatively, can communicate electrical power (e.g., when implementing the version with the controller **250A** that is housed in the handle housing **211**). The cable **121** includes a removable connection jack so that the handle assembly **210** can be readily discarded after a single use and the power adapter **120** can be reused with subsequent handle assemblies. The fluid line connection **126** can include a luer fitting and a flow on-off valve so that a user can removably connect the handle assembly to a pole-mounted saline bag or other fluid source **125** without the need for an external pump mechanism positioned exterior to the handle housing **211**.

[0120] Referring to FIGS. 27-35, in operation of the system **200**, a user can translate the actuator **216** of the handle assembly **210** in the actuator slot **213** so as to retract and extend the abrasive elements **240a-c** from the distal end of the sheath **232**. For example, the retraction of the abrasive elements **240a-c** within the sheath **232** can advantageously cover the abrasive surfaces during navigation of abrasive elements **240a-c** to the target area, thereby reducing the likelihood of scraping or engaging unintended regions of a vessel along the navigation path. The user can then selectively adjust the abrasive elements **240a-c** distally from the sheath **232** lumen after navigating to the targeted site (e.g., for rotational atherectomy treatment). Also as described in more detail below, the system **200** can be configured to automatically prevent of rotation of the drive shaft **236** when the abrasive elements **240a-c** are retracted within the sheath lumen (FIGS. 28-29) and to provide selective activation of rotation of the drive shaft when the abrasive elements are in an extended position (e.g., FIGS. 30-31 and 32-33). Accordingly, the system **200** can provide an added safety control that facilitates improved and intuitive operation for a user both during navigation and

during rotational atherectomy treatment.

[0121] In some embodiments, the user can grasp the actuator **216** to urge movement along the actuator slot **213**, which causes the internal carriage assembly **245** to slidably translate along the longitudinal axis of the handle assembly **210**, as indicated by the arrow **215** to various positions within the actuator slot **213**. As the carriage assembly **245** is translated in relation to the housing **211**, the drive shaft **236** translates in relation to the sheath **232** in a corresponding manner. As such, the user can retract and extend the distal portion of the drive shaft **236** (including the abrasive elements **240a-c**) relative to the distal end of the sheath **232**. For example, the user can translate the actuator **216** proximally to the position of FIGS. **28**, **29**, and **35** to retract the abrasive elements **240a-c** into the sheath **232**.

[0122] In the proximal position depicted in FIGS. **28**, **29**, and **35**, the abrasive elements **240a-c** are retracted and housed within the distal end of the sheath **232**. The sheath **232** covering the abrasive elements **240a-c** facilitates navigation to and from a targeted small artery, including those where the vessel interior diameter is often 3 mm or less and the access path follows a tortuous route. For example, sheath **232** can include a smooth outer surface that facilitates ease of navigation and protection of the tortuous route that is not the targeted area (e.g., prevents the abrasive elements **240a-c** from engaging with areas outside of the target area).

[0123] Responsive to translation of the actuator **216** into the proximal position depicted in FIGS. **28**, **29**, and **35**, power is not supplied to the electrical motor **212** regardless of whether the rotational power button of the actuator **216** is actuated. For example, the handle assembly **210** can include a switch **241** (FIG. **35**) that controls power delivery to the electrical motor **212**. In some embodiments, the switch **241** is at least one of a mechanical switch, an optical switch, a magnetic switch, a rotary switch, a push button, or combinations thereof. In the proximal position, the carriage assembly **245** connected to the actuator **216** slides proximally and contacts a switch arm **243** to depresses the switch **241**. With the switch **241** depressed, power is not delivered to the electrical motor **212** and rotation of the drive shaft **236** is prevented. As such, with the abrasive elements **240a-c** retracted within the sheath **232**, the drive shaft **236** and abrasive elements **240a-c** are prevented from rotation while the abrasive elements **240a-c** are retracted within the sheath **232**. The prevention of rotation of the drive shaft **236** and abrasive elements **240a-c** (which can be executed by the controller described above) advantageously reduce the likelihood one or more abrasive elements **240a-c** from abrading an interior of the sheath **232**. Accordingly, in the proximal position, if a user presses the power button, power will not be supplied to the electric motor **212**, and the drive shaft **236** will not rotate.

[0124] In the threshold position depicted in FIGS. **30**, **31**, and **34**, the abrasive elements **240a-c** are extended out of the distal end of the sheath **232** to a minimum distance from the distal end of the sheath **232** where the abrasive elements **240a-c** are permitted to rotate and revolve in an orbital path to thereby contact and remove portions of a target lesion **207** (even those portions of the lesion that are spaced farther from the axis of the drive shaft **236** than the maximum radius of the abrasive elements **240a-c**).

[0125] Still referring to FIGS. **27-35**, responsive to translation of the actuator **216** into the threshold position depicted in FIGS. **30**, **30**, and **34**, power is supplied to the electrical motor **212**. For example, in the threshold position, the carriage assembly **245** connected to the actuator **216** slides distally and releases contact with the switch arm **243** and disengages the switch **241**. With the switch **241** disengaged, power is delivered to the electrical motor **212** and rotation of the drive shaft **236** is permitted when the user presses the power button. Accordingly, in the threshold position, responsive to a user pressing the power button, power is supplied to the electric motor **212**, and the drive shaft **236** rotates.

[0126] In the distal position depicted in FIGS. **32** and **33**, the abrasive elements **240a-c** are extended further from the distal end of the sheath **232** to a maximum distance from the distal end of the sheath **232** where the abrasive elements **240a-c** are permitted to rotate and revolve in an orbital

path to thereby contact and remove portions of a target lesion **207** (even those portions of the lesion that are spaced farther from the axis of the drive shaft **236** than the maximum radius of the abrasive elements **240a-c**).

[0127] Responsive to translation of the actuator **216** into the distal position depicted in FIGS. **30**, **30**, and **34**, the electrical motor **212** remains connected to power. For example, in the distal position, the carriage assembly **245** connected to the actuator **216** slides further distally and the switch arm **243** remains disengaged from the switch **241**. With the switch **241** disengaged, power is delivered to the electrical motor **212** and rotation of the drive shaft **236** is permitted when the user presses the power button. Accordingly, in the distal position, responsive to a user pressing the power button, power is supplied to the electric motor **212**, and the drive shaft **236** rotates. In some embodiments, contemporaneous with the rotation of the drive shaft **236**, the drive shaft **236** can be translated back and forth (distally and proximally) along the longitudinal axis of the drive shaft **236** between the threshold position and the distal position. Hence, the stenotic lesion **207** can be abraded radially and longitudinally by virtue of the simultaneous translation and orbital rotation of the abrasive elements **240a-c**. If the user translates the actuator **216** proximally past the threshold position, power to the electric motor **212** is stopped, and rotation of the abrasive elements **240a-c** also stops.

[0128] To operate the handle assembly **210** during a rotational atherectomy procedure, a clinician can grasp the actuator **216**, actuate (translate) the actuator **216** to a position at or distal of the threshold position shown in FIGS. **30**, **31**, and **34** and depress rotational power button (on the actuator **216**) with the same hand. During the actuation or translation of the actuator **216** distally (e.g., from the proximal position shown in FIGS. **28**, **29**, and **35**), the abrasive elements **240a-c** are advanced distally out of the distal end of the sheath **232** and are exposed to the target area. Additionally, the switch **241** is disengaged and power is supplied to the electric motor **217**. The clinician can move (translate) the actuator **116** along the slot **213** distally and proximally by hand (e.g., back and forth in relation to the housing **211**), while maintaining the rotational power button of the actuator **216** in the depressed state and while maintaining the abrasive elements **240a-c** at a position beyond the distal end of the sheath **232**. In that manner, a target lesion **207** can be abraded radially and longitudinally by virtue of the resulting orbital rotation and translation of the abrasive elements **240a-c**.

[0129] To further operate the handle assembly **210** during a rotational atherectomy procedure, a clinician can select a rotational speed using electrical switches **217a** and **217b**. In some cases, the rotational speed can be selected through a set of predefined speeds (e.g., at least two predefined speed settings, such as “low” and “high”) with electrical switch **217a** causing an increase in the speed setting and electrical switch **217b** causing a decrease in the speed setting. Optionally, each of the electrical switches **217a-b** can also include a light indicator. For example, when the electrical switches **217a-b** allow for selection for a “high” and “low” speed, respectively, the electrical switches **217a-b** can each have a single light, such that when a speed is selected, the light corresponding to the selected electrical switch **217a** or **217b** is illuminated to inform a clinician of the selected speed. In some embodiments, the light can shine through electrical switches **217** and **217b**. Alternatively, a light can be positioned proximal electrical switch **217a-b**. As another example, when the electrical switches **217a-b** allow modification of a speed between a range of speeds, the light indicator can be a light bar, such that a number of lights illuminated on the light bar correspond to a selected speed.

[0130] Still referring to FIGS. **27-35**, to further operate the handle assembly **210**, the handle assembly **210** can include a fluid pump switch **217c**. The fluid pump switch **217c** can share the features of fluid pump switch **117c** such as activating the internal fluid pump **214** to draw fluid (e.g., saline in this embodiment) from the fluid line **126** and urge the fluid through the sheath **232** toward the distal end portion of the drive shaft **236**. As such, the fluid pump switch **217c** can be used to both initially prime the sheath **232** (and remove air before insertion into the patient) and

then selectively activate additional flush fluid through the sheath **232** and into the vessel **205**. In some cases, a first depression of the fluid pump switch **217c** will turn the internal pump **214** on, while a second depression will turn the pump **214** off. In some embodiments, the fluid pump switch **217c** includes a light indicator, such that when the pump is on, a light is illuminated to inform the clinician that the pump is on.

[0131] In the depicted embodiment, the handle assembly **210** also includes a guidewire brake **218** that can share features with the guidewire brake **118**. For example, the guidewire brake **218** can be selectively actuated (e.g., pivoted relative to the handle housing **211** in this embodiment) to releasably clamp the guidewire **234** in a stationary position relative to the handle assembly **210** (and, in turn, stationary in relation to rotations of the drive shaft **236** during an atherectomy treatment). While the drive shaft **236** and handle assembly **210** are being advanced over the guidewire **234** to put the one or more abrasive elements **238** into a targeted position within a patient's vessel, the guidewire brake **218** is in a non-activated state (e.g., pivoted counter-clockwise about the central guidewire axis) from a rear perspective) so that the handle assembly **210** is free to slide in relation to the guidewire **234**. Then, when the clinician is ready to begin the atherectomy treatment, the guidewire brake **218** can be activated (e.g., pivoted clockwise about the central guidewire axis) to mechanically engage an exterior of the guidewire **234** and thereby releasably detain/lock the guidewire **234** in relation to the handle assembly **210**. That way, the guidewire **234** will not rotate while the drive shaft **236** is rotating, and the guidewire **234** will not translate while the actuator **216** is being manually translated in the direction **215**.

[0132] In some embodiments, the handle assembly **110** can include a guidewire brake light **219** that positioned along an upper face of the handle housing **211** at a position proximal to the other user interface buttons **217a-c** and adjacent to the guidewire brake **218**. As such, a user can readily view the guidewire brake light **219** (e.g., similar to the guidewire brake light **119**) and receive confirmation of whether the guidewire brake **218** is fully activated (to clamp the guidewire **234**) before selecting the rotational speed (e.g., buttons **217a-b**) and activating rotation (e.g., button on the actuator **216**). As such, the screenless user interface of the handle assembly **210** can provide a simplified and fluid hand motion for the user while also communicating effective information to the user.

[0133] Optionally, the controller **250** (or **250A** in other embodiments) can be configured to prevent the electric motor **212** from driving rotation of the drive shaft **236** until: (1) the guidewire brake **218** is activated (e.g., with the guidewire brake light **119** illuminated), (2) the pump **214** is activated to drive the flush fluid (e.g., via actuation of fluid pump switch **117c** that then illuminates the button **117c**), (3) a rotation speed has been selected via speed selection switches **217a** and **217b** (e.g., with a speed indicator light thereon being activated), (4) when the switch **241** is disengaged in response to the position of the actuator **216**, or a combination of all these conditions. As another example, the indicator lights associated with the selection switches **217a** and **217b**, the fluid pump switch **217c**, and the guidewire brake light **219** will alert a clinician that the rotational atherectomy system **200** should not be operated until all four systems (the position of the actuator, the motor, the pump, the guidewire brake) are activated and in position for use. For example, each system may have a green light, such that three green lights indicates the clinician can proceed with the atherectomy procedure. Optionally, only (1) the switch **241** is disengaged and (2) the guidewire **118** is actuated to allow rotation of the rotational atherectomy system **200**. Optionally, only the switch **241** is disengaged to allow rotation of the rotational atherectomy system **200**.

[0134] Still referring to FIGS. **27-35**, the rotational atherectomy system **200** also includes the controller **150**, which in this embodiment includes a processor and computer-readable memory storing control instructions thereon. The controller **150** is configured to receive input from sensors housed within the handle assembly, to receive input from the user interface on the handle assembly **210** (e.g., switches/actuators **216**, **217a-c**, **218**, and **241**), and to control the activation of the electric motor **212** and the pump **214** (responsive to inputs at the user interface switches/actuators). In this

embodiment, the controller **150** is contained inside the housing **122** of the power adapter **120** so that it is reusable with subsequent handle assemblies after the first handle assembly **210** is discarded (e.g., after use with a first patient). As previously described, the cable **121** can provide data communication between the controller **150** and the components of the user interface (e.g., switches/actuators **216**, **217a-c**, **218**, and **241**), the electric motor **212**, the pump **214**, and the feedback sensors housed within the handle assembly **210**. In an alternative embodiment, the controller (including the processor and computer-readable memory storing the control instructions) can be provided in the form of controller **250A** configured to be contained within the housing **211** of the handle assembly **210** (in proximity to the electric motor **212** and the pump **214**). In both options, the handle assembly **210** can be operated by a clinician using the above-described simplified, screenless interface to perform and control the rotational atherectomy procedure (e.g., without a user interface display screen along the handle assembly or on the units connected to the handle assembly). Preferably, the controller **150** (or controller **250A**) is housed in a manner that is sealed from fluids encountered by the handle assembly, such as saline, blood, or others.

[0135] A number of embodiments of the invention have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the invention. For example, design features of the embodiments described herein can be combined with other design features of other embodiments described herein. Accordingly, other embodiments are within the scope of the following claims.

Claims

1. A rotational atherectomy system for slidable insertion through a 4-French introducer sheath, comprising: a 4-French introducer rotational atherectomy device, comprising: a torque-transmitting coil of one or more filars that are helically wound around in a filar wind direction from a distal end to a proximal end to define an outer coil diameter and a drive shaft axis; and a series of abrasive eccentric burrs fixedly mounted to a distal end portion of the torque-transmitting coil and having a maximum burr diameter of no greater than 1.25 mm such that a burr-to-coil diameter ratio defined by each respective burr diameter and the outer coil diameter is 1.7 or less, wherein the series of abrasive eccentric burrs includes a distal eccentric abrasive burr, an intermediate eccentric abrasive burr, and a proximal eccentric abrasive burr that are equally sized and mounted in an eccentric spiral arrangement along the torque-transmitting coil so that centers of mass for the distal, intermediate, and proximal abrasive eccentric burrs are offset from the drive shaft axis in different planes; and a rotational atherectomy handle assembly coupled to a proximal end of the torque-transmitting coil and housing an electric motor configured to, responsive to user input at an actuator of the rotational atherectomy handle assembly, drive rotation of the series of abrasive burrs about the drive shaft axis in a rotational drive direction.
2. The rotational atherectomy system claim 1, further comprising a sheath that extends from the rotational atherectomy handle assembly and over the torque transmitting coil, wherein the rotational atherectomy handle assembly is configured to longitudinally adjust the torque-transmitting coil relative to the sheath such that, in a proximal position, the series of abrasive burrs are housed within a distal-most end of the sheath.
3. The rotational atherectomy system of claim 2, wherein responsive to distal longitudinal movement of the actuator of the rotational atherectomy handle, the series of abrasive burrs are adjusted to a distal position so as to extend distally from the distal-most end of the sheath.
4. The rotational atherectomy system of claim 3, wherein the electric motor housed within the rotational atherectomy handle is prevented from rotating the torque-transmitting coil when the series of abrasive burrs are in the proximal position and housed within the distal-most end of the sheath.
5. The rotational atherectomy system of claim 4, wherein when the series of abrasive burrs are

adjusted to the distal position to extend distally from the distal-most end of the sheath, a switch that is housed within the rotational atherectomy handle is triggered and the electric motor is selectively activatable to drive rotation of the series of abrasive burrs.

6. The rotational atherectomy system of claim 5, wherein when the actuator of the rotational atherectomy handle assembly is adjusted to a location proximal of a threshold position along an upper face of the rotational atherectomy handle assembly, the series of abrasive burrs are in the proximal position and housed within the distal-most end of the sheath.

7. The rotational atherectomy system of claim 6, wherein when the actuator of the rotational atherectomy handle assembly is adjusted to a location distal of the threshold position along the upper face of the rotational atherectomy handle assembly, the series of abrasive burrs are in the distal position and extend distally from the distal-most end of the sheath.

8. The rotational atherectomy device of claim 5, wherein contemporaneous with selective activation to drive rotation of the series of abrasive burrs, the torque transmitting coil is selectively translatable distally and proximally along the drive shaft axis.

9. The rotational atherectomy device of claim 8, further comprising a power button at the rotational atherectomy handle assembly that is selectable to activate the electric motor.

10. The rotational atherectomy device of claim 9, wherein, responsive to selection of the power button with the series of abrasive burrs in the distal position, the electric motor activates to drive rotation of the series of abrasive burrs.

11. The rotational atherectomy system of claim 1, wherein the distal abrasive eccentric burr has a center of mass offset in a first plane, and the intermediate abrasive eccentric burr has a center of mass offset in a second plane that is radially spaced from the first plane at the radial spacing angle of 20 degrees to 60 degrees.

12. The rotational atherectomy system of claim 11, wherein the proximal abrasive eccentric burr has a center of mass offset in a third plane that is radially spaced from the second plane at the radial spacing angle of 20 degrees to 60 degrees.

13. The rotational atherectomy system of claim 11, wherein a combined radial spacing angle for all abrasive eccentric burrs in the series of abrasive eccentric burrs is less than 90 degrees.

14. The rotational atherectomy system of claim 1, wherein each abrasive burr of the series of abrasive burrs is a metallic burr coated with diamond grit.

15. The rotational atherectomy system of claim 1, wherein said the electric motor housed within the rotational atherectomy handle is configured to rotate the torque-transmitting coil so that the series of abrasive burrs orbit in the rotational drive direction about the drive shaft axis.

16. The rotational atherectomy system of claim 15, wherein said filar wind direction from the distal end to the proximal end of torque-transmitting coil is opposite from the rotational drive direction urged by the electric motor housed within the rotational atherectomy handle.

17. The rotational atherectomy system of claim 1, wherein the torque transmitting coil is selectively translatable distally and proximally along the drive shaft axis while the electric motor drives rotation of the series of abrasive burrs.

18. The rotational atherectomy system of claim 17, further comprising a power button positioned on the actuator of the rotational atherectomy handle assembly and being selectable to activate the electric motor.

19. The rotational atherectomy system of claim 1, wherein the abrasive eccentric burrs are metallic components welded to the torque-transmitting coil.

20. The rotational atherectomy system of claim 1, wherein series of abrasive eccentric burrs consists of three equally sized spherical burrs fixedly mounted in the eccentric spiral arrangement along the torque-transmitting coil.
